

**PySAWIT:**  
**OIL PALM GROWTH AND YIELD MODEL**

Brief report (draft)

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# TABLE OF CONTENTS

|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| <b>TABLE OF CONTENTS .....</b>                                                | <b>II</b> |
| <b>CHAPTER 1. METEOROLOGY .....</b>                                           | <b>1</b>  |
| 1.1 Daily properties.....                                                     | 1         |
| 1.1.1 Rainfall .....                                                          | 1         |
| 1.1.2 Air temperature and solar radiation .....                               | 2         |
| 1.1.3 Wind speed .....                                                        | 2         |
| 1.1.4 Solar declination .....                                                 | 3         |
| 1.1.5 Sunrise, sunset, and day length.....                                    | 3         |
| 1.1.6 Solar constant .....                                                    | 3         |
| 1.1.7 Daily extraterrestrial solar irradiance .....                           | 4         |
| 1.1.8 Daily total solar irradiance and its direct and diffuse components .... | 4         |
| 1.2 Instantaneous (hourly) properties .....                                   | 4         |
| 1.2.1 Hour angle .....                                                        | 4         |
| 1.2.2 Solar position.....                                                     | 5         |
| 1.2.3 Extraterrestrial solar irradiance .....                                 | 5         |
| 1.2.4 Total solar irradiance and its direct and diffuse components .....      | 5         |
| 1.2.5 Air temperature.....                                                    | 6         |
| 1.2.6 Air vapour pressure .....                                               | 6         |
| 1.2.7 Wind speed .....                                                        | 7         |
| 1.2.8 Saturated air vapour pressure .....                                     | 7         |
| 1.2.9 Slope of the saturated vapour pressure curve .....                      | 8         |
| 1.2.10 Relative humidity (RH).....                                            | 8         |
| <b>CHAPTER 2. CANOPY PHOTOSYNTHESIS.....</b>                                  | <b>9</b>  |
| 2.1 PAR components within canopies .....                                      | 9         |
| 2.2 PAR absorption by the leaves.....                                         | 11        |
| 2.3 Sunlit and shaded leaf area index .....                                   | 11        |
| 2.4 Canopy temperature.....                                                   | 11        |
| 2.5 Temperature-dependent parameters.....                                     | 12        |
| 2.6 Gross leaf CO <sub>2</sub> assimilation.....                              | 13        |
| 2.7 Trend of change in CO <sub>2</sub> concentration .....                    | 15        |
| 2.8 Gross canopy CO <sub>2</sub> assimilation .....                           | 15        |
| 2.9 Daily integration .....                                                   | 15        |
| <b>CHAPTER 3. ENERGY BALANCE .....</b>                                        | <b>17</b> |
| 3.1 Net radiation .....                                                       | 17        |
| 3.2 Energy available to crop and soil.....                                    | 18        |
| 3.3 Soil/ground heat flux .....                                               | 18        |
| 3.4 Wind speed profile.....                                                   | 18        |
| 3.4.1 Crop roughness length and zero plane displacement .....                 | 18        |
| 3.4.2 Friction velocity.....                                                  | 19        |
| 3.4.3 Wind speed at canopy height.....                                        | 19        |
| 3.4.4 Eddy diffusivity and wind speed extinction coefficient.....             | 19        |
| 3.5 Flux resistances.....                                                     | 19        |
| 3.5.1 Aerodynamic resistances .....                                           | 19        |

|                                                 |                                                           |           |
|-------------------------------------------------|-----------------------------------------------------------|-----------|
| 3.5.1.1                                         | Soil surface to mean canopy flow .....                    | 19        |
| 3.5.1.2                                         | Mean canopy flow to reference level .....                 | 20        |
| 3.5.2                                           | Bulk boundary layer resistance .....                      | 20        |
| 3.5.3                                           | Stomatal conductance .....                                | 21        |
| 3.5.4                                           | Canopy resistance .....                                   | 22        |
| 3.5.5                                           | Soil surface resistance .....                             | 22        |
| 3.6                                             | Total latent heat flux .....                              | 23        |
| 3.7                                             | Air vapour pressure deficit at the mean canopy flow ..... | 24        |
| 3.8                                             | Latent heat fluxes for crop and soil .....                | 24        |
| 3.9                                             | Sensible heat fluxes for crop and soil .....              | 24        |
| 3.10                                            | Daily integration .....                                   | 25        |
| <b>CHAPTER 4. SOIL WATER .....</b>              |                                                           | <b>26</b> |
| 4.1                                             | Partitioning of soil profile into several layers .....    | 26        |
| 4.2                                             | Soil moisture characteristics .....                       | 26        |
| 4.3                                             | Soil layer depth and cumulative thickness .....           | 27        |
| 4.4                                             | Actual evaporation .....                                  | 28        |
| 4.5                                             | Actual transpiration .....                                | 28        |
| 4.6                                             | Root extraction of water .....                            | 29        |
| 4.7                                             | Saturated and unsaturated hydraulic conductivity .....    | 29        |
| 4.8                                             | Matric suction head .....                                 | 30        |
| 4.9                                             | Gravity head .....                                        | 30        |
| 4.10                                            | Hydraulic gradient .....                                  | 30        |
| 4.11                                            | Water content .....                                       | 30        |
| 4.12                                            | Net rainfall .....                                        | 32        |
| 4.13                                            | Daily integration .....                                   | 32        |
| <b>CHAPTER 5. MAINTENANCE RESPIRATION .....</b> |                                                           | <b>33</b> |
| 5.1                                             | Pinnae .....                                              | 33        |
| 5.2                                             | Rachis .....                                              | 33        |
| 5.3                                             | Trunk .....                                               | 33        |
| 5.4                                             | Roots .....                                               | 34        |
| 5.5                                             | Generative organs .....                                   | 34        |
| 5.6                                             | Total maintenance .....                                   | 34        |
| <b>CHAPTER 6. GROWTH RESPIRATION .....</b>      |                                                           | <b>36</b> |
| 6.1                                             | Annual VDM requirement .....                              | 36        |
| 6.2                                             | Daily VDM requirement .....                               | 36        |
| 6.3                                             | Dry matter partitioning .....                             | 36        |
| 6.4                                             | Growth rates .....                                        | 37        |
| 6.5                                             | Death rates of pinnae and rachis .....                    | 37        |
| 6.6                                             | Death rates of roots .....                                | 38        |
| 6.7                                             | Assimilates available for generative growth .....         | 38        |
| 6.8                                             | Plant part weights .....                                  | 38        |
| 6.9                                             | Leaf area index .....                                     | 39        |
| <b>CHAPTER 7. GENERATIVE ORGAN GROWTH .....</b> |                                                           | <b>40</b> |
| 7.1                                             | Boxcar train technique .....                              | 40        |
| 7.2                                             | Organ weights and yield .....                             | 42        |
| <b>CHAPTER 8. MISCELLANEOUS .....</b>           |                                                           | <b>44</b> |

|                                              |                                              |           |
|----------------------------------------------|----------------------------------------------|-----------|
| 8.1                                          | Trunk and tree height.....                   | 44        |
| 8.2                                          | Rooting depth.....                           | 44        |
| <b>REFERENCES.....</b>                       |                                              | <b>46</b> |
| <b>APPENDIX A. LIST OF MAIN SYMBOLS.....</b> |                                              | <b>50</b> |
| <b>APPENDIX B. GROUNDLEVEL OZONE .....</b>   |                                              | <b>60</b> |
|                                              | Daily concentration of ozone .....           | 60        |
|                                              | Trend of change in ozone concentration ..... | 60        |
|                                              | Instantaneous concentration of ozone .....   | 61        |
|                                              | Isoprene emission.....                       | 61        |
|                                              | Ozone created from isoprene emission .....   | 62        |
|                                              | Dry deposition velocity .....                | 62        |
|                                              | Net ozone concentration.....                 | 63        |
|                                              | Reduction of photosynthesis .....            | 64        |

## CHAPTER 1. METEOROLOGY

Calculations begin with the meteorology section. Properties such as solar irradiance, solar position, wind speed, air temperature and vapour pressure must be determined at the outset before calculating other factors that affect plant growth and yield.

### 1.1 Daily properties

Daily meteorological properties refer to meteorological properties for a given day of year (or date). For each day, daily values for solar radiation, rainfall (if any), relative humidity, minimum and maximum air temperature, and wind speed must be calculated. The stochastic weather generator WGEN by Richardson (1981) is modified to generate daily values of rainfall (if any) and minimum and maximum air temperature.

#### 1.1.1 Rainfall

The distribution of daily rainfall amounts is described using the two-parameter gamma distribution, given as:

$$f[r] = \frac{r^{\alpha-1} \exp(-r/\beta)}{\beta^\alpha \Gamma[\alpha]}$$

where  $f[r]$  is the probability density function for the rainfall amount  $r$  (mm);  $\alpha$  and  $\beta$  are the shape and scale distribution parameters, respectively; and  $\Gamma[\alpha]$  is the gamma function. The cumulative distribution function (CDF) of rainfall can be represented as

$$F[r] = \int_0^r f[x] dx$$

so that for a wet day, its amount of rainfall is determined by taking the inverse of the gamma distribution as

$$r = F^{-1}[1 - p]$$

where  $p$  is a uniformly distributed random probability value (0 to 1). Note that  $F^{-1}[1 - p]$  is taken rather than  $F^{-1}[p]$  because large rainfall amounts are associated with smaller probabilities, whereas small rainfall amounts with larger probabilities. In other words, smaller rainfalls are more typical (*i.e.*, occur more frequently) than larger rainfalls.

In the model, 12 monthly values of  $(\alpha, \beta)$  must be provided; one set of  $(\alpha, \beta)$  for each month. For each month,  $(\alpha, \beta)$  can be estimated by

$$\alpha = \left( \frac{\text{mean rainfall for the month}}{\text{std. dev of rainfall for the month}} \right)^2$$

$$\beta = \frac{\text{mean rainfall for the month}}{\alpha}$$

Whether rain would occur for a given day depends on whether it had rained on the previous day. A first-order Markov chain is used to generate the occurrence of wet and dry days. For each month, a set of  $[P(W|W), P(W|D)]$  must be provided, where  $P(W|W)$  is the probability of rain (W) on a given day (*e.g.*, today) when it had rained the previous day (*e.g.*, yesterday). Similarly,  $P(W|D)$  is the probability of rain on a given day when it had been dry (D) the previous day.

### 1.1.2 Air temperature and solar radiation

WGEN was originally calibrated against 20 years of weather data from 31 locations in the US. To adapt WGEN for more local conditions, the values for  $A$  and  $B$  matrices (Eq. 10 and 11 in Richardson, 1981) were modified to:

$$A = \begin{bmatrix} 0.420 & 0.114 & 0.039 \\ 0.017 & 0.622 & 0.004 \\ 0.279 & -0.022 & 0.183 \end{bmatrix}$$

$$B = \begin{bmatrix} -0.569 & 0.375 & 0.567 \\ 0.533 & -0.164 & 0.549 \\ -0.129 & 0.888 & 0.121 \end{bmatrix}$$

$A$  and  $B$  matrices were obtained using the procedure as described by Richardson (1981) and by using 1985 to 2007 daily weather data from Serdang. Nonetheless, air temperature simulations using these new  $A$  and  $B$  matrices still produced very similar results as using the original matrices.

Another modification to the WGEN model was the period in the year when daily minimum and maximum air temperature peaked. In the original WGEN model, this occurred around DOY (day of year) 200 (mid year) for both daily minimum and maximum air temperatures. However, examining the daily weather data for Serdang, Merlimau, and several other sites showed that this was not true. Generally, in Malaysia, daily minimum air temperature peaked at about DOY 120 and daily maximum air temperature at about DOY 75. Consequently, these two values were used instead in the modified WGEN.

Finally, the following must be provided to the model to generate daily minimum and maximum air temperature and solar radiation:

- a) Minimum air temperature (°C): mean annual value for dry days, amplitude (mean difference between the daily and mean air temperature), CV (coefficient of variation), amplitude for the CV, and the mean annual value for wet days.
- b) Maximum air temperature (°C): the same as for minimum air temperature.

### 1.1.3 Wind speed

Wind speed is described by the Weibull probability density function:

$$f[u_d] = \frac{k}{c} \left( \frac{u_d}{c} \right)^{k-1} \exp \left[ - \left( \frac{u_d}{c} \right)^k \right]$$

where  $f[u_d]$  is the probability density function for the daily wind speed  $u$  ( $\text{m s}^{-1}$ ), and  $k$  and  $c$  are the shape (unitless) and scale ( $\text{m s}^{-1}$ ) distribution parameters, respectively. The cumulative distribution function (CDF) of daily windspeed can be represented as

$$F[u_d] = 1 - \exp\left[-\left(\frac{u_d}{c}\right)^k\right]$$

so that the wind speed for a given day is determined by taking the inverse of the cumulative probability distribution as

$$u_d = c[-\ln(p)]^{1/k}$$

where  $p$  is a uniformly distributed random probability value (0 to 1).

In the model, 12 monthly values of  $(k, c)$  must be provided to generate the daily wind speed for any day in a given month. These  $(k, c)$  data set can be obtained from literature.

#### 1.1.4 Solar declination

$$\delta = -0.4093 \cos\left[\frac{2\pi}{365}(t_d + 10)\right]$$

where  $\delta$  is the solar declination (radians); and  $t_d$  is the day of year (Goudriaan and van Laar, 1994).

#### 1.1.5 Sunrise, sunset, and day length

$$\begin{aligned} t_{sr} &= 12 - \frac{12}{\pi} \arccos\left(-\frac{\sin \delta \cdot \sin \lambda}{\cos \delta \cdot \cos \lambda}\right) = 24 - t_{ss} \\ t_{ss} &= 12 + \frac{12}{\pi} \arccos\left(-\frac{\sin \delta \cdot \sin \lambda}{\cos \delta \cdot \cos \lambda}\right) \\ DL &= \frac{24}{\pi} \arccos\left(-\frac{\sin \delta \cdot \sin \lambda}{\cos \delta \cdot \cos \lambda}\right) = t_{ss} - t_{sr} \end{aligned}$$

where  $t_{sr}$ ,  $t_{ss}$  and  $DL$  are the sunrise, sunset and day length, respectively (all expressed in unit hour);  $\delta$  is the solar declination (radians); and  $\lambda$  is the site latitude (radians) (Goudriaan and van Laar, 1994).

#### 1.1.6 Solar constant

$$I_c = 1370 \times \left\{ 1 + 0.033 \cos\left[2\pi(t_d - 10) / 365\right] \right\}$$

where  $I_c$  is the solar constant ( $\text{W m}^{-2}$ ); and  $t_d$  is the day of year (Goudriaan and van Laar, 1994).

### 1.1.7 Daily extraterrestrial solar irradiance

$$I_{et,d} = 3600 \times I_c \times \frac{24}{\pi} \left[ a \times \arccos\left(-\frac{a}{b}\right) + b \sqrt{1 - \left(\frac{a}{b}\right)^2} \right]$$

$$a = \sin \lambda \sin \delta$$

$$b = \cos \lambda \cos \delta$$

where  $I_{et,d}$  is the daily extraterrestrial solar irradiance ( $\text{J m}^{-2} \text{ day}^{-1}$ );  $I_c$  is the solar constant ( $\text{W m}^{-2}$ );  $\delta$  is the solar declination (radians); and  $\lambda$  is the site latitude (radians) (Goudriaan and van Laar, 1994).

### 1.1.8 Daily total solar irradiance and its direct and diffuse components

$$I_{t,d} = \int_{t_{sr}}^{t_{ss}} I_t dt$$

$$I_{dr,d} = \int_{t_{sr}}^{t_{ss}} I_{dr} dt$$

$$I_{df,d} = \int_{t_{sr}}^{t_{ss}} I_{df} dt$$

where  $I_{t,d}$  and  $I_t$  are the daily ( $\text{J m}^{-2} \text{ day}^{-1}$ ) and instantaneous ( $\text{W m}^{-2}$ ) total solar irradiance, respectively;  $I_{df,d}$  and  $I_{df}$  are the daily ( $\text{J m}^{-2} \text{ day}^{-1}$ ) and instantaneous ( $\text{W m}^{-2}$ ) diffuse solar irradiance, respectively; and  $I_{dr,d}$  and  $I_{dr}$  are the daily ( $\text{J m}^{-2} \text{ day}^{-1}$ ) and instantaneous ( $\text{W m}^{-2}$ ) direct solar irradiance, respectively. These daily solar components are obtained by integrating their respective instantaneous values ( $I_t$ ,  $I_{df}$ , or  $I_{dr}$ ) over the period from sunrise ( $t_{sr}$ ) to sunset ( $t_{ss}$ ).

## 1.2 Instantaneous (hourly) properties

Instantaneous properties refer to meteorological properties for a given hour and day of year (or date and time).

### 1.2.1 Hour angle

$$\tau = \frac{\pi}{12} (t_h - 12)$$

where  $\tau$  is the hour angle (radians); and  $t_h$  is the local solar time (hour) (Goudriaan and van Laar, 1994).

*Note:* hour angles will be negative for before solar noon ( $t_h < 12$ ), positive for after solar noon ( $t_h > 12$ ), and zero at exactly solar noon.



### 1.2.2 Solar position

#### **Solar inclination (and solar height)**

$$\theta = \arccos(\sin \delta \sin \lambda + \cos \delta \cos \lambda \cos \tau)$$

where  $\theta$  is the solar inclination (angle from vertical) (radians);  $\delta$  is the solar declination (radians);  $\tau$  is the hour angle (hour); and  $\lambda$  is the site latitude (radians) (Goudriaan and van Laar, 1994).

*Note:* Rather than its angle from vertical, solar angle can be specified alternatively as  $\beta$ , the solar height or solar elevation, which is the solar angle from horizontal, determined by  $\pi/2 - \theta$  (radians).

#### **Solar azimuth**

$$\phi = \begin{cases} \pi - \arccos\left(\frac{\sin \lambda \sin \beta - \sin \delta}{\cos \lambda \cos \beta}\right) & t_h \leq 12 \\ \pi + \arccos\left(\frac{\sin \lambda \sin \beta - \sin \delta}{\cos \lambda \cos \beta}\right) & t_h > 12 \end{cases}$$

where  $\phi$  is the solar azimuth (angle from North in a clockwise direction) (radians);  $\delta$  is the solar declination (radians);  $\tau$  is the hour angle (hour); and  $\lambda$  is the site latitude (radians) (Goudriaan and van Laar, 1994).

### 1.2.3 Extraterrestrial solar irradiance

$$I_{et} = I_c \sin \beta = I_c \cos \theta$$

where  $I_{et}$  is the instantaneous extraterrestrial solar irradiance ( $\text{W m}^{-2}$ );  $I_c$  is the solar constant ( $\text{W m}^{-2}$ );  $\theta$  is the solar inclination (radians); and  $\beta$  is the solar height (or solar elevation) (radians) (Goudriaan and van Laar, 1994).

*Note:*  $\beta = \pi/2 - \theta$ .

### 1.2.4 Total solar irradiance and its direct and diffuse components

$$I_t = I_{dr} + I_{df}$$

where  $I_t$  is the instantaneous total solar irradiance ( $\text{W m}^{-2}$ );  $I_{dr}$  is the instantaneous direct solar irradiance ( $\text{W m}^{-2}$ ); and  $I_{df}$  is the instantaneous diffuse solar irradiance ( $\text{W m}^{-2}$ ).

The partitioning of  $I_t$  into  $I_{df}$  and  $I_{dr}$  is based on the empirical equation developed by Khatib et al. (2012) who used long-term daily solar radiation data (1984 to 2004) from 28 sites in Malaysia (including Sabah and Sarawak):

$$\frac{I_{df}}{I_t} = 0.9505 + 0.91634K_T - 4.851K_T^2 + 3.2353K_T^3$$

where  $K_T$  is the sky clearness index, which is simply:

$$K_T = \frac{I_t}{I_{et}}$$

where  $I_{et}$  is the instantaneous extraterrestrial solar irradiance ( $\text{W m}^{-2}$ ). The determination of the sky clearness index was determined by adapting the method by Dimas et al. (2011) by regressing  $K_T$  against  $RH$  (relative humidity) to obtain the following relationship:

$$K_T = 1.1857 - 0.0112RH$$

This equation was obtained by fitting the best function to the relationship between measured  $RH$  and  $K_T$  for five oil palm estates in Malaysia: Bukit Selarong, Kedah (5.462824 °N, 100.597084 °E); Diamond Jubilee, Melaka (2.33333 °N, 102.483333 °E); Imam, Sabah (4.333333 °N, 117.833333 °E); Seri Intan, Perak (3.976583 °N, 100.9739 °E); and Ulu Remis, Johor (1.827778 °N, 103.461944 °E) for the period between 2015 and 2016.

### 1.2.5 Air temperature

It is assumed that air temperature varies sinusously with time in the period between 1.5 hours after the time of minimum air temperature and time of sunset. For time outside this period, air temperature is assumed to change linearly with time.

$$T_a = \begin{cases} T_{set} + \frac{(T_{min} - T_{set})(24 + t_h - t_{ss})}{(t_{sr} + 1.5) + (24 - t_{ss})} & t_h < (t_{sr} + 1.5) \\ T_{min} + (T_{max} - T_{min}) \sin \left[ \frac{\pi(t_h - t_{sr} - 1.5)}{t_{ss} - t_{sr}} \right] & (t_{sr} + 1.5) \leq t_h \leq t_{ss} \\ T_{set} + \frac{(T_{min} - T_{set})(t_h - t_{ss})}{(t_{sr} + 1.5) + (24 - t_{ss})} & t_h > t_{ss} \end{cases}$$

$$T_{set} = T_{min} + (T_{max} - T_{min}) \sin \left[ \frac{\pi(t_{ss} - t_{sr} - 1.5)}{t_{ss} - t_{sr}} \right]$$

where  $T_a$  is the air temperature (°C) at local solar time  $t_h$  (hour);  $T_{min}$  and  $T_{max}$  are the minimum and maximum air temperature (°C), respectively; and  $t_{sr}$  and  $t_{ss}$  are the times of sunrise and sunset, respectively (hour); and  $T_{set}$  is the air temperature (°C) at sunset ( $t_{ss}$ ).

### 1.2.6 Air vapour pressure

$$e_a = 6.1078 \exp \left( \frac{17.269 T_{dew,cal}}{T_{dew,cal} + 237.3} \right)$$

where  $e_a$  is the air vapour pressure (mbar); and  $T_{dew,cal}$  is the calibrated dew point temperature (°C), and it should be lower than current air temperature,  $T_a$ :

$$T_{dew,cal} = MIN(T_a, T_{dew})$$

where  $T_{dew}$  is the average dew point temperature for the site (mean for Malaysia, 23 °C).

### 1.2.7 Wind speed

The instantaneous values for wind speed vary sinuously within the day as

$$u = MAX \left\{ u_{min}, u_{min} + (u_{max} - u_{min}) \sin \left[ \frac{\pi}{DL} (t_h - t_{sr} - 1.5) \right] \right\}$$

where  $u$  is the wind speed for the given hour ( $m\ s^{-1}$ );  $u_{max}$  and  $u_{min}$  are the maximum and minimum wind speed for the day, respectively ( $m\ s^{-1}$ );  $t_h$  is the local solar hour (hour);  $t_{sr}$  and  $t_{ss}$  are the hour of sunrise and sunset, respectively (hour); and  $DL$  is the day length (hour).

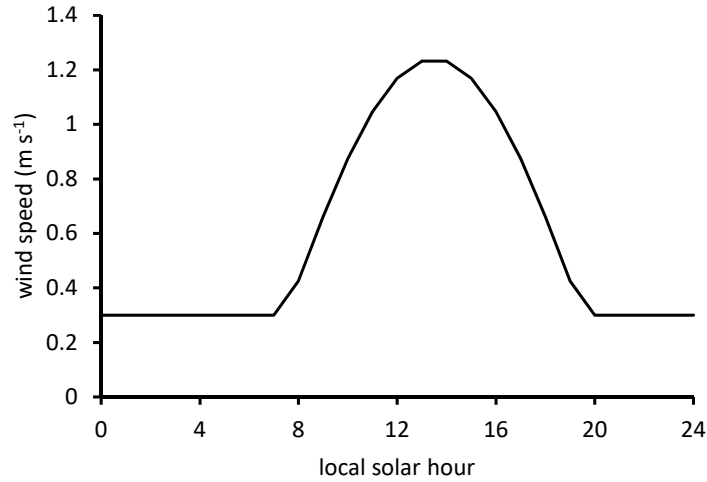


Fig. 1.1. Idealized distribution of wind speed in a day

The mean daily minimum and maximum wind speed can be estimated by the following empirical equations, developed from measured data from several towns in Malaysia:

$$u_{min} = 0.5591u_d^{1.25}$$

$$u_{max} = 1.7976u_d^{0.75}$$

where  $u_d$  is the mean daily wind speed ( $m\ s^{-1}$ ).

### 1.2.8 Saturated air vapour pressure

$$e_s \llbracket T_a \rrbracket = 6.1078 \exp \left[ \frac{17.269T_a}{T_a + 237.3} \right]$$

where  $e_s[[T_a]]$  is the function returning the saturated vapour pressure (mbar) at air temperature  $T_a$  (°C).

### 1.2.9 Slope of the saturated vapour pressure curve

$$\Delta = \frac{25029.4 \exp[17.269T_a/(T_a + 237.3)]}{(T_a + 237.3)^2}$$

where  $\Delta$  is the slope of the saturated vapour pressure curve (mbar K<sup>-1</sup> or equivalently, mbar °C<sup>-1</sup>) at air temperature  $T_a$  (°C).

### 1.2.10 Relative humidity (RH)

$$RH = 100 \frac{e_a}{e_s[[T_a]]}$$

where  $RH$  is the relative humidity (%) for the given hour;  $e_a$  is the air vapour pressure (mbar); and  $e_s[[T_a]]$  is the function returning the saturated vapour pressure (mbar) at air temperature  $T_a$  (°C).

## CHAPTER 2. CANOPY PHOTOSYNTHESIS

This section describes the amount of CO<sub>2</sub> assimilated by the canopies. The method by Collatz *et al.* (1991) was adapted. Essentially, calculations involved determining the solar irradiance within the canopies, calculating the CO<sub>2</sub> assimilation rate by individual leaves, then scaling up to canopy assimilation rate.

### 2.1 PAR components within canopies

There are four flux components within the canopies, and they are determined from Campbell and Norman (1998) as

$$Q_{p,dr} = (1 - p_{dr})Q_{dr} \exp(-k_{dr}\omega\sqrt{\alpha}L)$$

$$Q_{p,dr,dr} = (1 - p_{dr})Q_{dr} \exp(-k_{dr}\omega L)$$

$$Q_{p,dr,\alpha} = \frac{(Q_{p,dr} - Q_{p,dr,dr})}{2}$$

$$\bar{Q}_{p,df} = \frac{(1 - p_{df})Q_{df} [1 - \exp(-k_{df}\sqrt{\alpha}L)]}{k_{df}\sqrt{\alpha}L}$$

#### *Canopies flux components*

$Q_{p,dr}$  is the PAR irradiance for unintercepted beam *with* scattered beam ( $\mu\text{mol photons m}^{-2}$  ground  $\text{s}^{-1}$ );  $Q_{p,dr,dr}$  is the PAR irradiance for unintercepted beam *without* scattering ( $\mu\text{mol photons m}^{-2}$  ground  $\text{s}^{-1}$ );  $Q_{p,dr,\alpha}$  is the PAR irradiance of the scattered component *only* ( $\mu\text{mol photons m}^{-2}$  leaf  $\text{s}^{-1}$ ); and  $\bar{Q}_{p,df}$  is the mean diffuse irradiance ( $\mu\text{mol photons m}^{-2}$  leaf  $\text{s}^{-1}$ ).

*Note:* Both  $Q_{p,dr,\alpha}$  and  $\bar{Q}_{p,df}$  are expressed based on a per unit leaf area (not per unit ground area). It is assumed that scattered and diffuse irradiance on a unit horizontal ground area is equal to their irradiance on a unit leaf area.

#### *PAR direct and diffuse irradiance*

$$Q_{dr} = I_{dr} \times 0.5 \times 4.55$$

$$Q_{df} = I_{df} \times 0.5 \times 4.55$$

where  $Q_{dr}$  and  $Q_{df}$  are, respectively, the instantaneous direct and diffuse PAR irradiance, both expressed in  $\mu\text{mol photons m}^{-2}$  ground  $\text{s}^{-1}$ ; and  $I_{dr}$  and  $I_{df}$  are the instantaneous direct and diffuse solar irradiance, respectively ( $\text{W m}^{-2}$  ground).

*Note:* Total PAR irradiance is assumed 50% of total solar irradiance. It is then converted from unit  $\text{W m}^{-2}$  ground to  $\mu\text{mol photons m}^{-2}$  ground  $\text{s}^{-1}$  by multiplying it with 4.55 (Goudriaan and van Laar, 1994).

### ***Canopy extinction coefficient for direct PAR***

$$k_{dr} = \frac{0.5}{\sin \beta} = \frac{0.5}{\cos \theta}$$

where  $k_{dr}$  is the canopy extinction coefficient for direct PAR;  $\beta$  is the solar height (or solar elevation) (radians); and  $\theta$  is the solar inclination ( $\beta = \pi/2 - \theta$ ) (radians) (Goudriaan, 1977).

*Note:* random distribution of leaves in the canopies is assumed.

### ***Canopy clustering coefficient (correcting for non-uniform distribution of canopies in the aerial space)***

$$\omega = \omega_0 + 6.6557(1 - \omega_0) \exp[-\exp(-\theta + 2.2103)]$$
$$\omega_0 = -\frac{1}{k_{dr}L} \ln \left\{ \tau_b + (1 - \tau_b) \exp \left[ -k_{dr} \frac{L}{(1 - \tau_b)} \right] \right\}$$

where  $\omega$  is the canopy clustering coefficient (0 to 1);  $\omega_0$  is the canopy clustering coefficient (0 to 1) when the sun is at zenith (highest point in the sky);  $L$  is the total leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground);  $\theta$  is the solar inclination (radians); and  $k_{dr}$  is the canopy extinction coefficient for direct PAR (Kustas and Norman, 1999).  $\tau_b$  is the canopy gap fraction (0 to 1) determined from Awal (2008) as

$$\tau_b = (1 + 1.33\sqrt{L})^{-1}$$

where  $L$  is the total leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground).

### ***Canopy extinction coefficient for diffuse PAR***

$$k_{df} = \exp(0.038042 - 0.38845\sqrt{L})$$

where  $k_{df}$  is the canopy extinction coefficient for diffuse PAR; and  $L$  is the total leaf area index (LAI) ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground) (adapted from Teh, 2006).

Note that the equation for  $k_{df}$  already accounts for canopy gaps and that its assumption, like for  $k_{dr}$ , is the canopy is also randomly distributed in the aerial space.

### ***Canopy reflection and leaf absorption coefficients for PAR***

The canopy reflection coefficient for direct ( $p_{dr}$ ) and diffuse ( $p_{df}$ ) PAR are determined by

$$p_{dr} = \text{MAX} \left[ 0.04, p_s \exp(-2k_{dr}\omega\sqrt{\alpha}L) \right]$$
$$p_{df} = \text{MAX} \left[ 0.04, p_s \exp(-2k_{df}\sqrt{\alpha}L) \right]$$

where  $k_{dr}$  and  $k_{df}$  are the canopy extinction coefficients for direct and diffuse PAR, respectively;  $\omega$  is the canopy clustering coefficient (0 to 1);  $L$  is the total leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground);  $\alpha$  is the leaf scattering coefficient for PAR, taken as 0.8 (Goudriaan, 1977; Goudriaan and van Laar, 1994); and  $p_s$  is the soil reflection coefficient, taken as 0.15.

## 2.2 PAR absorption by the leaves

$$Q_{sl} = \alpha (k_{dr}\omega Q_{dr} + \bar{Q}_{p,df} + Q_{p,dr,\alpha})$$

$$Q_{sh} = \alpha (\bar{Q}_{p,df} + Q_{p,dr,\alpha})$$

where  $Q_{sl}$  and  $Q_{sh}$  are the PAR flux densities absorbed by the sunlit leaves and shaded leaves, respectively ( $\mu\text{mol photons m}^{-2}$  leaf  $\text{s}^{-1}$ );  $k_{dr}$  is the canopy extinction coefficient for direct PAR;  $\omega$  is the canopy clustering coefficient (0 to 1);  $Q_{dr}$  is the instantaneous direct PAR irradiance ( $\mu\text{mol photons m}^{-2}$  ground  $\text{s}^{-1}$ );  $Q_{p,dr,\alpha}$  is the PAR irradiance of the scattered component *only* ( $\mu\text{mol photons m}^{-2}$  leaf  $\text{s}^{-1}$ );  $\bar{Q}_{p,df}$  is the mean diffuse irradiance ( $\mu\text{mol photons m}^{-2}$  leaf  $\text{s}^{-1}$ ); and  $\alpha$  is the leaf scattering coefficient (where in this case, it is also the leaf absorption coefficient) for PAR (0.8) (Campbell and Norman, 1998).

*Note:* PAR flux densities absorbed ( $Q_{sl}$  and  $Q_{sh}$ ) are based on a per unit leaf area (not per unit ground area).

## 2.3 Sunlit and shaded leaf area index

$$L_{sl} = \frac{1 - \exp(-k_{dr}\omega L)}{k_{dr}\omega}$$

$$L_{sh} = L - L_{sl}$$

where  $L_{sl}$  and  $L_{sh}$  are sunlit and shaded leaf area index, respectively ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground);  $L$  is the total leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground);  $k_{dr}$  is the canopy extinction coefficient for direct PAR; and  $\omega$  is the canopy clustering coefficient (0 to 1) (Goudriaan and van Laar, 1994).

## 2.4 Canopy temperature

$$T_f = \frac{H_c r_a^c + (H_s + H_c) r_a^a}{\rho c_p} + T_r$$

where  $T_f$  is the canopy (foliage) temperature ( $^{\circ}\text{C}$ );  $T_r$  is the air temperature at reference level ( $^{\circ}\text{C}$ );  $H_s$  and  $H_c$  are the soil and crop sensible heat fluxes, respectively ( $\text{W m}^{-2}$ );  $r_a^a$  is the aerodynamic resistance between the mean canopy flow and reference level ( $\text{s m}^{-1}$ );  $r_a^c$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ ); and  $\rho c_p$  is the volumetric heat capacity for air ( $1221.09 \text{ J m}^{-3} \text{ K}^{-1}$ ) (Shuttleworth and Wallace, 1985).

## 2.5 Temperature-dependent parameters

There are some parameters whose values are sensitive to the canopy/foilage temperature  $T_f$ . Consequently, their values are adjusted according to

$$\xi = \xi_{(25)} \times Q_{10,\xi}^{(T_f-25)/10}$$

where  $\xi_{(25)}$  is the parameter value at 25 °C ( $K_c(25)$ ,  $K_o(25)$ ,  $\tau(25)$  or  $V_{c,max(25)}$ );  $T_f$  is the canopy temperature (°C); and  $Q_{10,\xi}$  is the relative change in the parameter  $\xi$  for every 10 °C change. Values for the model parameters at 25 °C and their respective  $Q_{10}$  values are given below.

Table 2.1. Parameter values for the photosynthesis calculations. Parameters with subscript (25) denote their values at 25 °C. Except for  $V_{c,max}$ , all values are from Bernacchi *et al.* (2001) and Bernacchi *et al.* (2002) for C3 plants.

| $\xi_{(25)}$    | Description                                         | Value  | Unit                                 | $Q_{10,\xi}$ |
|-----------------|-----------------------------------------------------|--------|--------------------------------------|--------------|
| $K_c(25)$       | Michaelis-Menten constant for CO <sub>2</sub>       | 270    | μmol mol <sup>-1</sup>               | 2.786        |
| $K_o(25)$       | Michaelis-Menten constant for O <sub>2</sub>        | 165000 | μmol mol <sup>-1</sup>               | 1.355        |
| $\tau(25)$      | CO <sub>2</sub> / O <sub>2</sub> specificity factor | 2800   | μmol μmol <sup>-1</sup>              | 0.703        |
| $V_{c,max(25)}$ | Maximum Rubisco capacity rate                       | 79     | μmol m <sup>-2</sup> s <sup>-1</sup> | 2.573        |

The maximum Rubisco capacity rate for oil palm declines with tree age, from 91.1 μmol m<sup>-2</sup> s<sup>-1</sup> at year 1 to 69.8 μmol m<sup>-2</sup> s<sup>-1</sup> at year 19. The relationship between  $V_{c,max(25)}$  at oil palm tree age can be described by

$$V_{c,max(25)} = 87.935 - 0.0026age$$

where *age* is the tree age (days).

### **Michaelis-Menten constant for CO<sub>2</sub>**

$$K_c = K_c(25) \times 2.786^{(T_f-25)/10}$$

where  $K_c$  is the Michaelis-Menten constant for CO<sub>2</sub> (μmol mol<sup>-1</sup>); and  $T_f$  is the canopy temperature (°C)

### **Michaelis-Menten constant for O<sub>2</sub>**

$$K_o = K_o(25) \times 1.355^{(T_f-25)/10}$$

where  $K_o$  is the Michaelis-Menten constant for CO<sub>2</sub> (μmol mol<sup>-1</sup>); and  $T_f$  is the canopy temperature (°C)



### **CO<sub>2</sub>/O<sub>2</sub> specificity factor**

$$\tau = \tau_{(25)} \times 0.703^{(T_f - 25)/10}$$

where  $\tau$  is the CO<sub>2</sub>/O<sub>2</sub> specificity factor (unitless); and  $T_f$  is the canopy temperature (°C)

### **CO<sub>2</sub> compensation point**

$$\Gamma^* = \frac{O_a}{2\tau}$$

where  $\Gamma^*$  is the CO<sub>2</sub> compensation point (μmol mol<sup>-1</sup>);  $O_a$  is the ambient O<sub>2</sub> concentration in air (210000 μmol mol<sup>-1</sup>); and  $\tau$  is CO<sub>2</sub>/O<sub>2</sub> specificity factor (μmol μmol<sup>-1</sup>) (Collatz *et al.*, 1991).

### **Rubisco maximum capacity rate**

$V_{c,max}$  is the Rubisco maximum capacity rate (μmol CO<sub>2</sub> m<sup>-2</sup> leaf s<sup>-1</sup>), and its value changes according to temperature by:

$$V_{c,max} = \frac{V_{c,max(25)} \times 2.573^{(T_f - 25)/10}}{1 + \exp[0.29(T_f - 40)]}$$

where  $T_f$  is the canopy temperature (°C).

*Note:*  $V_{c,max}$  is corrected additionally for temperatures greater than 40 °C after which the temperature causes a rapid decline in the capacity rate due to Rubisco degradation.

## **2.6 Gross leaf CO<sub>2</sub> assimilation**

$$\Lambda_{sl/sh} = \text{MIN}(v_c, v_{q,sl/sh}, v_s)$$

where  $\Lambda_{sl/sh}$  is the gross leaf CO<sub>2</sub> assimilation (μmol CO<sub>2</sub> m<sup>-2</sup> leaf s<sup>-1</sup>) for either sunlit (subscript *sl*) or shaded (subscript *sh*) leaves; and  $\text{MIN}()$  is a function to select the minimum of the three assimilation rates ( $v_c$ ,  $v_{q,sl/sh}$  and  $v_s$ ) (Collatz *et al.*, 1991).

### **Rubisco-limited assimilation ( $v_c$ )**

$$v_c = \frac{V_{c,max} (C_i - \Gamma^*)}{K_c (1 + O_a/K_o) + C_i}$$

where  $v_c$  is the Rubisco-limited leaf CO<sub>2</sub> assimilation rate (μmol CO<sub>2</sub> m<sup>-2</sup> leaf s<sup>-1</sup>);  $K_c$  is the Michaelis-Menten constant for CO<sub>2</sub> (μmol mol<sup>-1</sup>);  $K_o$  is the Michaelis-Menten constant for O<sub>2</sub> (μmol mol<sup>-1</sup>);  $\Gamma^*$  is the CO<sub>2</sub> compensation point (μmol mol<sup>-1</sup>);  $O_a$  is the ambient O<sub>2</sub>

concentration in air (210000  $\mu\text{mol mol}^{-1}$ );  $C_i$  is the intercellular  $\text{CO}_2$  concentration ( $\mu\text{mol mol}^{-1}$ ); and  $V_{c,\max}$  is the Rubisco capacity rate ( $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ ) (Collatz *et al.*, 1991).

*Note:*  $V_{c,\max}$ ,  $\Gamma^*$ ,  $K_c$  and  $K_o$  must be corrected for canopy temperature.

#### **Light-limited assimilation ( $v_q$ )**

$$v_{q,sl/sh} = e_m \alpha Q_{sl/sh} \frac{C_i - \Gamma^*}{C_i + 2\Gamma^*}$$

where  $v_{q,sl/sh}$  is the light-limited leaf  $\text{CO}_2$  assimilation rate ( $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ leaf s}^{-1}$ ) for either sunlit (subscript  $sl$ ) or shaded (subscript  $sh$ ) leaves;  $\Gamma^*$  is the  $\text{CO}_2$  compensation point ( $\mu\text{mol mol}^{-1}$ );  $C_i$  is the intercellular  $\text{CO}_2$  concentration ( $\mu\text{mol mol}^{-1}$ );  $Q_{sl/sh}$  is the PAR absorbed by either sunlit (subscript  $sl$ ) or shaded (subscript  $sh$ ) leaves ( $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ );  $\alpha$  is the leaf absorption of PAR (0.8); and  $e_m$  is the intrinsic quantum efficiency or quantum yield ( $\mu\text{mol CO}_2 \mu\text{mol}^{-1} \text{ photons}$ ) (Collatz *et al.*, 1991). The value for  $e_m$  is taken as 0.051.

#### **Sink-limited assimilation ( $v_s$ )**

$$v_s = 0.5V_{c,\max}$$

where  $v_s$  is the sink-limited leaf  $\text{CO}_2$  assimilation rate ( $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ leaf s}^{-1}$ ); and  $V_{c,\max}$  is the Rubisco capacity rate ( $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ ) (Collatz *et al.*, 1991).

*Note:*  $V_{c,\max}$  must be corrected for canopy temperature.

#### **Intercellular $\text{CO}_2$ concentration**

The relationship between intercellular  $\text{CO}_2$  concentration ( $C_i$ ) to ambient  $\text{CO}_2$  concentration ( $C_a$ ) is described following Yin and van Laar (2005) as

$$\frac{C_i}{C_a} = 1 - \left( 1 - \frac{\Gamma^*}{C_a} \right) (a + bD_{leaf})$$

where  $C_i$  and  $C_a$  are the intercellular and ambient  $\text{CO}_2$  concentration, respectively ( $\mu\text{mol mol}^{-1}$ );  $\Gamma^*$  is the  $\text{CO}_2$  compensation point ( $\mu\text{mol mol}^{-1}$ ).  $D_{leaf}$  is the leaf vapour pressure deficit (mbar) which is determined by

$$D_{leaf} = e_s \left[ T_f \right] - e_a$$

where  $e_s(T_f)$  is the function returning the saturated vapour pressure (mbar) at foliage temperature  $T_f$  ( $^{\circ}\text{C}$ ); and  $e_a$  is the air vapour pressure (mbar).

Leaf measurements on various ages of oil palm tree (from 1 to 19 years) revealed that empirical coefficients  $a$  and  $b$  are 0.0615 and 0.0213, respectively.

## 2.7 Trend of change in CO<sub>2</sub> concentration

Ambient CO<sub>2</sub> concentration is assumed to change per day in a linear rate using the following equation:

$$C_{a,td2} = C_{a,td1} + \frac{r_{\Delta Ca}}{365}$$

where  $C_{a,(td1)}$  and  $C_{a,(td2)}$  are the ambient CO<sub>2</sub> concentration at day of ear  $td1$  and  $td2$ , respectively; and  $r_{\Delta Ca}$  is the annual rate of change in ambient CO<sub>2</sub> concentration ( $\mu\text{mol mol}^{-1} \text{yr}^{-1}$ ). The parameter  $r_{\Delta Ca}$  is taken as a constant, which means that the rate of change follows a linear trend.

## 2.8 Gross canopy CO<sub>2</sub> assimilation

$$\Lambda_{canopy} = \Lambda_{sl}L_{sl} + \Lambda_{sh}L_{sh}$$

where  $\Lambda_{canopy}$  is the gross canopy CO<sub>2</sub> assimilation ( $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ ground s}^{-1}$ );  $\Lambda_{sl}$  and  $\Lambda_{sh}$  are the gross leaf CO<sub>2</sub> assimilation rates for sunlit and shaded leaves, respectively ( $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ leaf s}^{-1}$ ); and  $L_{sl}$  and  $L_{sh}$  are the sunlit and shaded leaf area index, respectively ( $\text{m}^2 \text{ leaf m}^{-2} \text{ ground}$ ) (Campbell and Norman, 1998).

## 2.9 Daily integration

The daily gross canopy photosynthesis is determined by

$$\Lambda'_{canopy,d} = 3600 \times 30 \times 10^{-9} \times \frac{10000}{PD} \times \int_{t_{sr}}^{t_{ss}} \Lambda_{canopy} dt'$$

where  $\Lambda'_{canopy,d}$  is the daily gross photosynthesis (converted to  $\text{kg CH}_2\text{O palm}^{-1} \text{ day}^{-1}$ );  $\Lambda_{canopy}$  is the gross canopy CO<sub>2</sub> assimilation for the hour ( $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ ground s}^{-1}$ );  $PD$  is the planting density ( $\text{palms ha}^{-1}$ ); and  $t_{sr}$  and  $t_{ss}$  are the sunrise and sunset hour (hour).

Numerical method Gaussian integration is used to obtain the daily gross canopy photosynthesis. Five points ( $N=5$ ) over the diurnal period (from sunrise to sunset) are selected, and for each selected hour, the gross canopy CO<sub>2</sub> assimilation is calculated (Goudriaan and van Laar, 1994).

Note: to account for ozone effects, gross photosynthesis could be reduced further and linearly by

$$\Lambda''_{canopy,d} = \Lambda'_{canopy,d} \times (1 - f_{O3})$$

where  $\Lambda''_{canopy,d}$  and  $\Lambda'_{canopy,d}$  are the canopy photosynthesis with and without ozone effects, respectively ( $\text{kg CH}_2\text{O palm}^{-1} \text{ day}^{-1}$ ); and  $f_{O3}$  is the reduction in assimilates due to ozone (fraction).



## CHAPTER 3. ENERGY BALANCE

The energy balance of the soil-plant-atmosphere system is described as a network of resistances in which heat fluxes must traverse within the system (Shuttleworth and Wallace, 1985). Solving the energy balance gives the water loss by soil evaporation and plant transpiration.

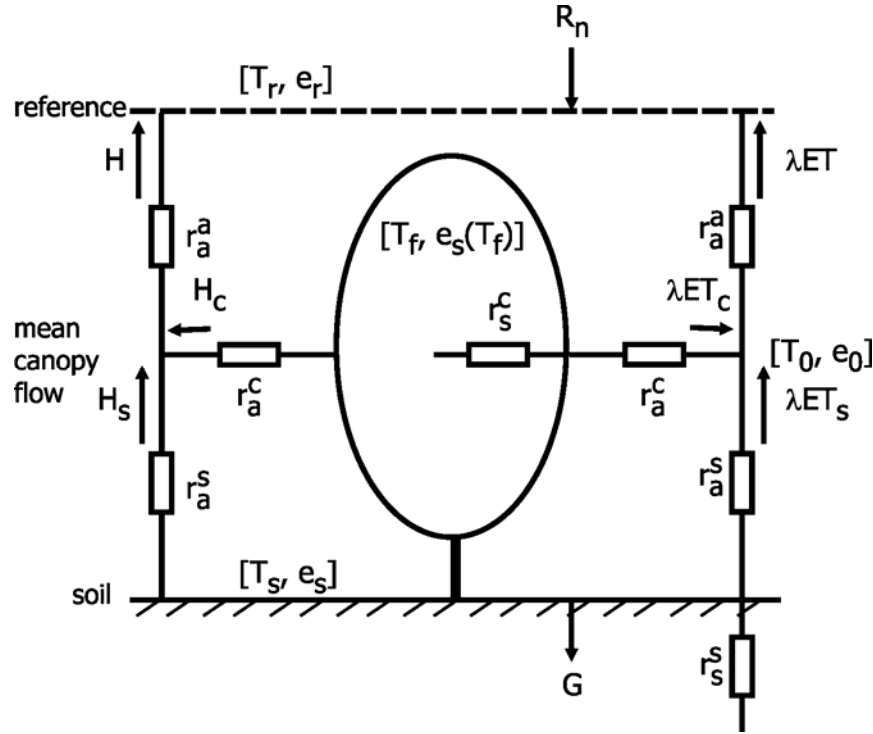


Fig. 3.1 Energy balance described as a network of resistances. Key:  $\lambda ET$  and  $\lambda ET_c$  and  $\lambda ET_s$  are the latent heat fluxes from the system, crop and soil, respectively;  $H$ ,  $H_c$  and  $H_s$  are the sensible heat fluxes from the system, crop and soil, respectively;  $R_n$  is the net radiation flux into the system;  $G$  is the heat conduction into the soil;  $T_r$ ,  $T_f$ ,  $T_o$  and  $T_s$  are the temperatures for the reference height, crop, mean canopy flow and soil, respectively;  $e_r$ ,  $e_s$  and  $e_o$  are the vapour pressure at the reference height, soil surface and mean canopy flow, respectively;  $e_s(T)$  is saturated vapour pressure at temperature  $T$ ;  $r_a^a$  is the aerodynamic resistance between the mean canopy flow and reference height;  $r_a^s$  is the aerodynamic resistance between the soil and mean canopy flow;  $r_a^c$  is the bulk boundary layer resistance;  $r_s^c$  and  $r_s^s$  are the canopy and soil surface resistance, respectively.

### 3.1 Net radiation

$$R_n = (1 - p)I_t + R_{nL}$$

$$R_{nL} = 0.98\sigma(T_a + 273.15)^4 \left[ 1.31 \left( \frac{e_a}{T_a + 273.15} \right)^{1/7} - 1 \right]$$

where  $R_n$  is the net radiation ( $\text{W m}^{-2}$ );  $R_{nL}$  is the net longwave radiation ( $\text{W m}^{-2}$ );  $I_t$  is the instantaneous total solar irradiance ( $\text{W m}^{-2}$ );  $p$  is the surface albedo (typically, 0.15);  $T_a$  is the air temperature ( $^{\circ}\text{C}$ );  $e_a$  is the air vapour pressure (mbar); and  $\sigma$  is the Stefan-Boltzmann constant ( $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ ).

### 3.2 Energy available to crop and soil

$$A = A_c + A_s = R_n - G$$

$$A_c = (1 - \tau_{dr,\alpha}) R_n$$

$$A_s = \tau_{dr,\alpha} R_n - G$$

$$\tau_{dr,\alpha} = \exp(-k_{dr}\omega\sqrt{\alpha}L)$$

where  $A$  is the net radiation available to the system (soil and crop);  $A_c$  is the fraction of net radiation available to the crop ( $\text{W m}^{-2}$ );  $A_s$  is the fraction of net radiation available to the soil ( $\text{W m}^{-2}$ );  $G$  is the soil heat flux ( $\text{W m}^{-2}$ );  $L$  is the total leaf area index (LAI);  $k_{dr}$  is the canopy extinction coefficient for direct solar radiation;  $\omega$  is the canopy clustering coefficient (0 to 1); and  $\alpha$  is the scattering coefficient for total solar radiation (0.5 from Goudriaan and val Laar, 1994).

### 3.3 Soil/ground heat flux

$$G = R_n [t_c + \tau_{dr,\alpha} (t_s - t_c)]$$

$$\tau_{dr,\alpha} = \exp(-k_{dr}\omega\sqrt{\alpha}L)$$

where  $G$  is the soil heat flux ( $\text{W m}^{-2}$ );  $L$  is total leaf area index (LAI);  $k_{dr}$  is the canopy extinction coefficient for direct solar radiation;  $\omega$  is the canopy clustering coefficient (0 to 1);  $\alpha$  is the scattering coefficient for total solar radiation (0.5);  $R_n$  is the net radiation ( $\text{W m}^{-2}$ ); and  $t_c$  and  $t_s$  are the fraction of net radiation as soil heat flux under full canopies and for bare soil (no canopies), respectively.  $t_c$  and  $t_s$  are taken as 0.05 (Monteith, 1973) and 0.315 (Kustas and Daughtry, 1990), respectively.

### 3.4 Wind speed profile

#### 3.4.1 Crop roughness length and zero plane displacement

The method by Massman (1997) is used to determine the zero plane displacement height  $d$  and crop roughness length  $z_0$ :

$$d = hA$$

$$z_0 = h(1 - A)\exp(-k/0.32)$$

$$A = 1 - \frac{\exp(-2\alpha)[\exp(-2\alpha) - 1]}{2\alpha} \quad 0.30 \leq A \leq 0.95$$

where  $d$  is the zero plane displacement height (m);  $z_0$  is the crop roughness length (m);  $\alpha$  is the vertical wind speed extinction coefficient;  $h$  is the plant height (m);  $k$  is the von Karman constant (0.4); and  $C_d$  is the foliage drag coefficient (0.32).

### 3.4.2 Friction velocity

$$u^* = \frac{ku}{\ln\left(\frac{z_r - d}{z_0}\right)}$$

where  $u^*$  is the friction velocity ( $\text{m s}^{-1}$ );  $k$  is the von Karman constant (0.4); and  $u$  is the wind speed ( $\text{m s}^{-1}$ ) at the given hour;  $z_r$  is at the reference height (m);  $d$  is the zero plane displacement height (m);  $z_0$  is the crop roughness length (m) (Shuttleworth and Wallace, 1985).

### 3.4.3 Wind speed at canopy height

$$u_h = \frac{u^*}{k} \ln\left(\frac{h - d}{z_0}\right)$$

where  $u_h$  is the wind speed function returning the wind speed ( $\text{m s}^{-1}$ ) at full tree or canopy height  $h$  (m);  $d$  is the zero plane displacement height (m);  $z_0$  is the crop roughness length (m);  $k$  is the von Karman constant (0.4);  $u^*$  is the friction velocity ( $\text{m s}^{-1}$ ) (Shuttleworth and Wallace, 1985).

### 3.4.4 Eddy diffusivity and wind speed extinction coefficient

Vertical wind speed extinction coefficient  $\alpha$  (assumed equal to eddy diffusivity  $n$ ) increases with increasing leaf area index  $L$ . Simulations by Massman (1987) for hypothetical canopies with uniform foliage density suggested that the relationship between  $\alpha$  can be approximated by the following function:

$$n = \alpha = 3[1 - \exp(-L)]$$

where  $L$  is the total leaf area index (LAI) ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground) (Nikolov and Zeller, 2003).

## 3.5 Flux resistances

The heat fluxes have to traverse various types of resistances within the soil-plant-atmosphere system. This section describes each of those resistance types.

### 3.5.1 Aerodynamic resistances

#### 3.5.1.1 Soil surface to mean canopy flow

$$r_a^s = \frac{\exp(n)}{nku^*} \left[ \exp\left(-n \frac{z_{s0}}{h}\right) - \exp\left(-n \frac{z_0 + d}{h}\right) \right]$$

where  $r_a^s$  is the resistance between the soil surface and the mean canopy flow ( $\text{s m}^{-1}$ );  $h$  is the plant height (m);  $k$  is the von Karman constant (0.4);  $u^*$  is the friction velocity ( $\text{m s}^{-1}$ );  $n$  is the eddy diffusivity extinction coefficient;  $d$  is the zero plane displacement height (m);  $z_0$  is the crop roughness length (m); and  $z_{s0}$  is the soil surface roughness length (m) (Shuttleworth and Wallace, 1985).

*Note:* For flat, tilled land,  $z_{s0}$  is taken as 0.004 m (Hansen, 1993).

### 3.5.1.2 Mean canopy flow to reference level

$$r_a^a = \frac{1}{ku^*} \ln \left( \frac{z_r - d}{h - d} \right) + \frac{1}{nku^*} \left\{ \exp \left[ n \left( 1 - \frac{z_0 + d}{h} \right) \right] - 1 \right\}$$

where  $r_a^a$  is the resistance between the mean canopy flow and the reference level ( $\text{s m}^{-1}$ );  $h$  is the plant height (m);  $k$  is the von Karman constant (0.4);  $u^*$  is the friction velocity ( $\text{m s}^{-1}$ );  $n$  is the eddy diffusivity extinction coefficient;  $d$  is the zero plane displacement height (m);  $z_0$  is the crop roughness length (m); and  $z_r$  is the reference height (m) (Shuttleworth and Wallace, 1985).

### 3.5.2 Bulk boundary layer resistance

$$r_a^c = \frac{\alpha}{0.01 L_{eff} [1 - \exp(-\alpha/2)] \sqrt{u_h/w}}$$

where  $r_a^c$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ );  $L_{eff}$  is the effective leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground);  $w$  is the mean leaf width (m);  $u_h$  is the function returning the wind speed at the canopy top (*i.e.*, at plant height  $h$ ) ( $\text{m s}^{-1}$ ); and  $\alpha$  is the wind speed extinction coefficient (Mitchell, 1976; Shuttleworth and Wallace, 1985).

Effective leaf area index is determined by

$$L_{eff} = \min \left( L, \frac{L_{\max, PD}}{2} \right)$$

where  $L$  is the leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground); and  $L_{\max, PD}$  is the maximum leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground) for a given  $PD$  planting density (palms  $\text{ha}^{-1}$ ) (Szeicz and Long, 1969) and it is calculated as:

$$L_{\max, PD} = 0.0274 PD^{\frac{1}{A}}$$

where  $A$  is 0.935.

The pinnae length and width can be determined by



$$l = 0.2191 \ln\left(\frac{age}{365}\right) + 0.475$$

$$w = 0.0152 \ln\left(\frac{age}{365}\right) + 0.0165$$

where  $l$  and  $w$  are the leaflet (pinnae) length and width, respectively (m); and  $age$  is the tree age (days). The equations for  $l$  and  $w$  were developed by fitting the best equation to the measured leaf dimension data from Rao et al. (1992).

### 3.5.3 Stomatal conductance

$$gst = gst_{max} \times f_{PAR} \times f_{water} \times f_D$$

where  $gst$  is the leaf stomatal conductance ( $\text{m s}^{-1}$ );  $gst_{max}$  is the maximum leaf stomatal conductance ( $500 \text{ mmol m}^{-2} \text{ s}^{-1}$  or  $0.012077 \text{ m s}^{-1}$  from Kallarackal *et al.*, 2004); and  $f_{PAR}$ ,  $f_{water}$ , and  $f_D$  are reductions, scaled from 0 to 1, to  $gst_{max}$  due to PAR irradiance, water stress, and vapour pressure deficit, respectively.

#### PAR function

$f_{PAR}$  is determined by

$$f_{PAR} = \frac{gst[PAR]}{gst[PAR_{max}]}$$

$$gst[PAR] = a[1 - \exp(-bPAR)]$$

$$a = 0.014614$$

$$b = 0.008740$$

where  $gst[PAR]$  is the function returning the leaf stomatal conductance ( $\text{m s}^{-1}$ ) at a given PAR irradiance ( $\text{W m}^{-2}$ ) ( $a$  and  $b$  values were obtained from leaf measurements of stomatal conductance for various oil palm tree ages at various PAR levels); and  $PAR_{max}$  is the maximum PAR irradiance ( $\text{W m}^{-2}$ ), taken as  $330 \text{ W m}^{-2}$  (light saturation at about  $1500 \mu\text{mol m}^{-2} \text{ s}^{-1}$  from Dufrêne and Saugier, 1993).

#### Water stress function

$f_{water}$  is determined as the ratio between actual and potential transpiration, or

$$f_{water} = \frac{T_a}{ET_c}$$

where  $T_a$  is the actual transpiration ( $\text{m day}^{-1}$ ); and  $ET_c$  is the potential transpiration ( $\text{m day}^{-1}$ ) (Kropff, 1993).

#### Vapour pressure deficit function

$f_D$  is determined by

$$f_D = \frac{gst[D]}{gst[D_{min}]}$$

$$gst[D] = a + b \ln(D)$$

$$a = 0.031970$$

$$b = -0.007516$$

where  $gst[D]$  is the function returning the leaf stomatal conductance ( $\text{m s}^{-1}$ ) at a given vapour pressure deficit,  $D$  (mbar), determined by

$$D = e_s[T_a] - e_a$$

where  $e_s[T_a]$  is the function returning the saturated vapour pressure (mbar) at air temperature  $T_a$  ( $^{\circ}\text{C}$ );  $e_a$  is the air vapour pressure (mbar); and  $D_{min}$  is the minimum vapour pressure deficit (mbar), taken as 10 mbar, based on measurements.

The coefficients  $a$  and  $b$  values were derived from fitting the best function to measured leaf stomatal conductance for various oil palm tree ages (year 1 to 19) at various vapour pressure deficits.

*Note:* Leaf stomatal conductance decreases with increasing vapour pressure deficit. Hence,  $D_{min}$  gives maximum leaf stomatal conductance.

### 3.5.4 Canopy resistance

Leaf stomatal conductance  $gst$  is determined, then scaled up to canopy resistance (inverse of canopy conductance) by the following equations:

$$r_s^c = \frac{1}{gst \times L_{eff}}$$

where  $r_s^c$  is the canopy resistance ( $\text{s m}^{-1}$ );  $L$  is the total leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground); and  $L_{eff}$  is the effective leaf area index ( $\text{m}^2$  leaf  $\text{m}^{-2}$  ground) (Szeicz and Long, 1969).

### 3.5.5 Soil surface resistance

The methods by Choudhury and Monteith (1988) and Farahani and Ahuja (1996) are used to determine the soil surface resistance as

$$r_s^s = r_{s,dry}^s \exp\left(-\frac{1}{\lambda} \times \frac{\theta_1}{\theta_{s,1}}\right)$$

$$r_{s,dry}^s = \frac{\tau l}{\phi_p D_{m,v}}$$

where  $r_s^s$  is the maximum soil surface resistance ( $\text{s m}^{-1}$ );  $D_{m,v}$  is the vapour diffusion coefficient in air ( $24.7 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ );  $\phi_p$  is the soil porosity (unitless);  $l$  is the dry soil layer

thickness (taken as the first soil layer thickness) (m);  $\theta_l$  is the soil water content of the first soil layer ( $\text{m}^3 \text{m}^{-3}$ );  $\theta_{s,1}$  is the saturated soil water content ( $\text{m}^3 \text{m}^{-3}$ ) in the first soil layer; and  $\lambda$  is the soil pore-size distribution index.  $\tau$  is the soil tortuosity (unitless) determined from Shen and Chen (2007) as

$$\tau = \sqrt{\phi_p + 3.79(1 - \phi_p)}$$

where  $\phi_p$  is the soil porosity (unitless).

*Note:* The soil profile is divided into two or more layers.

### 3.6 Total latent heat flux

$$\lambda ET = C_c PM_c + C_s PM_s$$

$$PM_c = \frac{\Delta A + (\rho c_p D - \Delta r_a^c A_s) / (r_a^a + r_a^c)}{\Delta + \gamma \left[ 1 + r_s^c / (r_a^a + r_a^c) \right]}$$

$$PM_s = \frac{\Delta A + [\rho c_p D - \Delta r_a^s A_c] / (r_a^a + r_a^s)}{\Delta + \gamma \left[ 1 + r_s^s / (r_a^a + r_a^s) \right]}$$

$$C_c = \left\{ 1 + R_c R_a / [R_s (R_c + R_a)] \right\}^{-1}$$

$$C_s = \left\{ 1 + R_s R_a / [R_c (R_s + R_a)] \right\}^{-1}$$

$$R_a = (\Delta + \gamma) r_a^a$$

$$R_c = (\Delta + \gamma) r_a^c + \gamma r_s^c$$

$$R_s = (\Delta + \gamma) r_a^s + \gamma r_s^s$$

where  $\lambda ET$  is the total latent heat flux ( $\text{W m}^{-2}$ );  $r_a^a$  is the aerodynamic resistance between the mean canopy flow and reference height ( $\text{s m}^{-1}$ );  $r_a^c$  is the aerodynamic resistance between the soil and mean canopy flow ( $\text{s m}^{-1}$ );  $r_a^s$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ );  $r_s^c$  and  $r_s^s$  are the canopy and soil surface resistance, respectively ( $\text{s m}^{-1}$ );  $A$ ,  $A_s$  and  $A_c$  are energy available to the system (total), soil and crop, respectively ( $\text{W m}^{-2}$ );  $\Delta$  is the slope of the saturated vapour pressure curve ( $\text{mbar K}^{-1}$ );  $\gamma$  is the psychrometric constant ( $0.658 \text{ mbar K}^{-1}$ );  $D$  is the vapour pressure deficit ( $\text{mbar}$ ); and  $\rho c_p$  is the volumetric heat capacity for air ( $1221.09 \text{ J m}^{-3} \text{K}^{-1}$ ) (Shuttleworth and Wallace, 1985).

*Note:* To convert latent heat fluxes from unit  $\text{W m}^{-2}$  (equivalent to  $\text{J m}^{-2} \text{s}^{-1}$ ) to depth of water per second ( $\text{mm s}^{-1}$ ), simply divide it by  $\lambda$ , the latent heat of vaporization of water ( $2454000 \text{ J kg}^{-1}$ ).

### 3.7 Air vapour pressure deficit at the mean canopy flow

$$D_0 = D + \frac{r_a^a}{\rho c_p} [\Delta A - (\Delta + \gamma) \lambda ET]$$

where  $D_0$  is the vapour pressure deficit at the mean canopy flow (mbar);  $\lambda ET$  is the total latent heat flux ( $\text{W m}^{-2}$ );  $r_a^a$  is the aerodynamic resistance between the mean canopy flow and reference height ( $\text{s m}^{-1}$ );  $A$  is the total energy available to the system ( $\text{W m}^{-2}$ );  $\Delta$  is the slope of the saturated vapour pressure curve ( $\text{mbar K}^{-1}$ );  $\gamma$  is the psychometric constant ( $0.658 \text{ mbar K}^{-1}$ );  $D$  is the vapour pressure deficit (mbar); and  $\rho c_p$  is the volumetric heat capacity for air ( $1221.09 \text{ J m}^{-3} \text{ K}^{-1}$ ) (Shuttleworth and Wallace, 1985).

### 3.8 Latent heat fluxes for crop and soil

$$\lambda ET_s = \frac{\Delta A_s + \rho c_p D_0 / r_a^s}{\Delta + \gamma (r_s^s + r_a^s) / r_a^s}$$

$$\lambda ET_c = \frac{\Delta A_c + \rho c_p D_0 / r_a^c}{\Delta + \gamma (r_s^c + r_a^c) / r_a^c}$$

where  $\lambda ET_s$  and  $\lambda ET_c$  are the soil and crop latent heat fluxes, respectively ( $\text{W m}^{-2}$ );  $A_s$  and  $A_c$  are energy available to the soil and crop, respectively ( $\text{W m}^{-2}$ );  $r_a^s$  is the aerodynamic resistance between the soil and mean canopy flow ( $\text{s m}^{-1}$ );  $r_a^c$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ );  $r_s^s$  and  $r_s^c$  are the canopy and soil surface resistance, respectively ( $\text{s m}^{-1}$ );  $\Delta$  is the slope of the saturated vapour pressure curve ( $\text{mbar K}^{-1}$ );  $\gamma$  is the psychometric constant ( $0.658 \text{ mbar K}^{-1}$ );  $D_0$  is the vapour pressure deficit at the mean canopy flow (mbar); and  $\rho c_p$  is the volumetric heat capacity for air ( $1221.09 \text{ J m}^{-3} \text{ K}^{-1}$ ) (Shuttleworth and Wallace, 1985).

### 3.9 Sensible heat fluxes for crop and soil

$$H_s = \frac{\gamma A_s (r_s^s + r_a^s) - \rho c_p D_0}{\Delta r_a^s + \gamma (r_s^s + r_a^s)}$$

$$H_c = \frac{\gamma A_c (r_s^c + r_a^c) - \rho c_p D_0}{\Delta r_a^c + \gamma (r_s^c + r_a^c)}$$

where  $H_s$  and  $H_c$  are the soil and crop sensible heat fluxes, respectively ( $\text{W m}^{-2}$ );  $A_s$  and  $A_c$  are energy available to the soil and crop, respectively ( $\text{W m}^{-2}$ );  $r_a^s$  is the aerodynamic resistance between the soil and mean canopy flow ( $\text{s m}^{-1}$ );  $r_a^c$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ );  $r_s^c$  and  $r_s^s$  are the canopy and soil surface resistance, respectively ( $\text{s m}^{-1}$ );  $\Delta$  is the slope of the saturated vapour pressure curve ( $\text{mbar K}^{-1}$ );  $\gamma$  is the psychrometric constant ( $0.658 \text{ mbar K}^{-1}$ );  $D_0$  is the vapour pressure deficit at the mean canopy flow, respectively ( $\text{mbar}$ ); and  $\rho c_p$  is the volumetric heat capacity for air ( $1221.09 \text{ J m}^{-3} \text{ K}^{-1}$ ) (Shuttleworth and Wallace, 1985).

### 3.10 Daily integration

Numerical method Gaussian integration is used to obtain the daily latent heat fluxes. This is so that the daily amount of water transpired (by plants) and evaporated (by soil) can be known. Five points ( $N=5$ ) over 24 hours in a day are selected, and for each selected hour, the heat fluxes are calculated (Goudriaan and van Laar, 1994).

## CHAPTER 4. SOIL WATER

For accurate simulation of the water content in the soil, the soil profile has to be divided into several layers, and for each layer, the soil water content is simulated based on Darcy's law.

### 4.1 Partitioning of soil profile into several layers

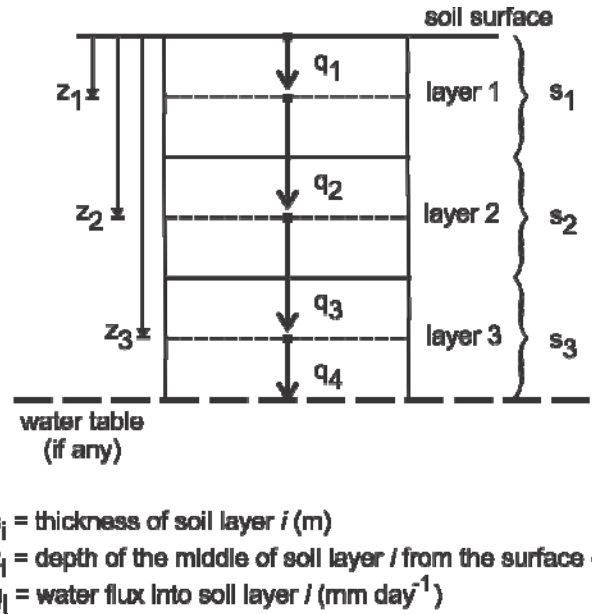


Fig. 4.1. Water flow in a soil profile which is divided into three successive layers, with the presence of a water table just beneath the third soil layer.

It is recommended to divide a soil profile into two or more consecutive layers, where soil layer  $i$  ( $i = 1$  to  $n$ ) has a thickness of  $s_i$  (m), and the depth from the soil surface to the middle of layer  $i$  is  $z_i$  (m). Water flux into soil layer  $i$  is denoted as  $q_i$  ( $\text{m day}^{-1}$ ).

#### Assumptions

1. No irrigation is supplied and no surface run-off and run-in occurs.
2. Water table, if any, has a constant depth from the soil surface.
3. Without the presence of a water table, the soil below the last soil layer is uniformly wet, and it has the same water content as the last soil layer. This means that water flows out of the last soil layer (*i.e.*,  $q_4$ ) due to gravitational pull alone (*i.e.*, no matric suction gradient).

### 4.2 Soil moisture characteristics

For each soil layer  $i$ , the volumetric soil water content at saturation, field capacity, and permanent wilting point are estimated based on clay, sand, and organic matter contents. Soil water characteristics (water content at saturation, field capacity, and permanent wilting points) are estimated based from Saxton and Rawls (2006).

### **Permanent wilting point**

$$\begin{aligned}\theta_{1500} &= \theta_{1500t} + (0.14\theta_{1500t} - 0.02) \\ \theta_{1500t} &= -0.024S + 0.487C + 0.006OM + 0.005(S \times OM) - 0.013(C \times OM) \\ &\quad + 0.068(S \times C) + 0.031\end{aligned}$$

where  $\theta_{1500}$  is the soil water content at permanent wilting point ( $\text{m}^3 \text{ m}^{-3}$ );  $S$  and  $C$  are the sand and clay contents, respectively (fraction); and  $OM$  is the organic matter content (%).

### **Field capacity**

$$\begin{aligned}\theta_{33} &= \theta_{33t} + (1.283\theta_{33t}^2 - 0.374\theta_{33t} - 0.015) \\ \theta_{33t} &= -0.251S + 0.195C + 0.011OM + 0.006(S \times OM) - 0.027(C \times OM) \\ &\quad + 0.452(S \times C) + 0.299\end{aligned}$$

where  $\theta_{33}$  is the soil water content at field capacity ( $\text{m}^3 \text{ m}^{-3}$ );  $S$  and  $C$  are the sand and clay contents, respectively (fraction); and  $OM$  is the organic matter content (%).

### **Saturation**

$$\begin{aligned}\theta_s &= \theta_{33} + \theta_{(S-33)} - 0.097S + 0.043 \\ \theta_{(S-33)} &= \theta_{(S-33)t} + 0.636\theta_{(S-33)t} - 0.107 \\ \theta_{(S-33)t} &= 0.278S + 0.034C + 0.022OM - 0.018(S \times OM) - 0.027(C \times OM) \\ &\quad - 0.584(S \times C) + 0.078\end{aligned}$$

where  $\theta_s$  is the soil water content at saturation ( $\text{m}^3 \text{ m}^{-3}$ );  $S$  and  $C$  are the sand and clay contents, respectively (fraction); and  $OM$  is the organic matter content (%).

## **4.3 Soil layer depth and cumulative thickness**

$$z_i = \sum_{j=1}^N 0.5(s_i + s_{i-1})$$

where  $z_i$  is the depth of soil layer  $i$  from the soil surface (m), taken from the middle of that soil layer to the soil surface;  $N$  is the number of soil layers (taken as 3);  $s_i$  and  $s_{i-1}$  are the thickness of soil layer  $i$  and  $i-1$ , respectively (m).

Note:  $s_0 = 0$ .

### **Cumulative thickness**

$$S_i = \sum_{j=1}^i s_j$$

where  $S_i$  is the cumulative thickness of soil layer  $i$  (m); that is, the sum of the thickness of layer  $i$  and all its preceding layers.

#### 4.4 Actual evaporation

$$E_a = ET_s \cdot R_{D,e}$$

$$R_{D,e} = \frac{1}{1 + \left[ 3.6073 \left( \theta_l / \theta_{s,1} \right) \right]^{-9.3172}}$$

where  $E_a$  is the actual evaporation (m day<sup>-1</sup>);  $ET_s$  is the potential evaporation (m day<sup>-1</sup>);  $R_{D,e}$  is the reduction factor for evaporation (ranging from 0 to 1) (van Keulen and Seligman, 1987); and  $\theta_l$  and  $\theta_{s,1}$  are the current and saturated soil water content in the first soil layer ( $i=1$ ), respectively (m<sup>3</sup> m<sup>-3</sup>).

*Note:* The soil water content from the first soil layer ( $i=1$ ),  $\theta_l$ , is taken because all evaporation occurs from this layer only (as this first layer is in contact with the atmosphere).

#### 4.5 Actual transpiration

$$T_a = ET_c \cdot R_{D,t}$$

$$R_{D,t} = \begin{cases} 1 & \theta_{root} \geq \theta_{cr,root} \\ \frac{\theta_{root} - \theta_{1500,root}}{\theta_{cr,root} - \theta_{1500,root}} & \theta_{1500,root} < \theta_{root} < \theta_{cr,root} \\ 0 & \theta_{root} \leq \theta_{1500,root} \end{cases}$$

$$\theta_{cr,root} = \theta_{1500,root} + 0.6(\theta_{s,root} - \theta_{1500,root})$$

where  $T_a$  is the actual transpiration (m day<sup>-1</sup>);  $ET_c$  is the potential transpiration (m day<sup>-1</sup>);  $R_{D,t}$  is the reduction factor for transpiration (ranging from 0 to 1) (Kropff, 1993);  $\theta_{root}$  is the total volumetric water content in the root zone (m<sup>3</sup> m<sup>-3</sup>);  $\theta_{1500,root}$  is the soil water content at wilting point (m<sup>3</sup> m<sup>-3</sup>) in the root zone;  $\theta_{s,root}$  is the soil water content at saturation (m<sup>3</sup> m<sup>-3</sup>) in the root zone; and  $\theta_{cr,root}$  is the volumetric water content in the root zone below which water stress occurs (m<sup>3</sup> m<sup>-3</sup>). Soil water measurements in irrigated and non-irrigated oil palm trials by Foong (1999) from 1983 to 1990 suggested that the critical soil water content below which oil palm suffers from water stress is approximately 60% of  $(\theta_{s,root} - \theta_{1500,root})$ . This critical point is equivalent to 45% of available water content (AWC) of Munchong soil series, the type of soil in Foong (1999). Rey *et al.* (1998) also observed that oil palm stomatal conductance would begin to decline when the soil water content fell below the level of 40 to 50% of AWC.

The algorithm to determine the total water content in the root zone,  $\theta_{root}$ , is as follows:



$$\theta_{root} = \frac{1}{d_{root}} \times \sum_{i=1}^N MAX[0, \theta_i(s_i - n_i)]$$

$$n_i = MAX(0, S_i - d_{root})$$

where  $d_{root}$  is the rooting depth (m);  $s_i$  is the thickness of soil layer  $i$  (m); and  $S_i$  is the cumulative thickness of soil layer  $i$  (m). The amount of water in the root zone is the summation of water content from the first soil layer ( $i=1$ ) until the rooting depth.

#### 4.6 Root extraction of water

The amount of water extracted by roots in each soil layer is based on the measured data for oil palm by Nelson *et al.* (2006) and on the water uptake algorithm by Miyazaki (2005):

$$T_{a,i} = T_a(\varphi_i - \varphi_{i-1})$$

$$\varphi_j = 1.8c_j - 0.8c_j^2$$

$$c_j = MIN\left(1, \frac{S_j}{d_{root}}\right)$$

where  $T_{a,i}$  is the amount of water extracted by roots in soil layer  $i$  (m day<sup>-1</sup>);  $d_{root}$  is the rooting depth (m);  $S_j$  is the cumulative thickness of soil layer  $j$  (m); and  $MIN()$  is the minimum of the enclosed values.

#### 4.7 Saturated and unsaturated hydraulic conductivity

The method by Bittelli *et al.* (2015) is followed. The saturated hydraulic conductivity in soil layer  $i$  ( $K_{si}$ , m day<sup>-1</sup>) is determined by

$$K_{s,i} = 864 \times 0.07 \times \left\{ \theta_{s,i} \left[ 1 - \left( \frac{\psi_e}{33} \right)^\lambda \right] \right\}^4$$

where  $\theta_{s,i}$  is the soil water content at saturation for soil layer  $i$  (m<sup>3</sup> m<sup>-3</sup>); and  $\lambda$  is slope of the logarithmic suction-soil moisture curve; and  $\psi_e$  is the air-entry suction (kPa), and they are determined by

$$\lambda = \frac{1}{B}$$

$$B = 8.25 - 1.26 \ln(d_g)$$

$$\psi_e = 3.9 - 0.61 \ln(d_g)$$

$$d_g = \exp[-1.96C + 2.3(1 - S - C) + 5.76S]$$

where  $d_g$  is the geometric mean distribution ( $\mu\text{m}$ ) of the soil's particles sizes: clay  $C$  and sand  $S$  (both in fractions).

The unsaturated hydraulic conductivity in soil layer  $i$  ( $k_i$ ,  $\text{m day}^{-1}$ ) is determined by

$$K_{\theta,i} = K_{s,i} \left( \frac{\theta_i}{\theta_{s,i}} \right)^{3+2/\lambda}$$

where  $K_{\theta,i}$  and  $K_{s,i}$  are the unsaturated and saturated hydraulic conductivity in soil layer  $i$ , respectively ( $\text{m day}^{-1}$ );  $\theta_i$  is the current volumetric soil water content in soil layer  $i$  ( $\text{m}^3 \text{ m}^{-3}$ );  $\theta_{s,i}$  is the soil water content at saturation ( $\text{m}^3 \text{ m}^{-3}$ ); and  $\lambda$  is the slope of the logarithmic suction-soil moisture curve, as determined previously.

#### 4.8 Matric suction head

The soil matric suction in soil layer  $i$  ( $H_{m,i}$ ,  $\text{m}$ ) is determined by

$$H_{m,i} = \begin{cases} 3.3 - \left[ \frac{(33 - \psi_e)(\theta_i - \theta_{33,i})}{10(\theta_{s,i} - \theta_{33,i})} \right] & \theta_i \geq \theta_{33,i} \\ \frac{\exp\left(\ln 33 + \frac{1}{\lambda} \ln \theta_{33,i}\right)}{10\theta_i^{1/\lambda}} & \theta_i < \theta_{33,i} \end{cases}$$

where  $H_{m,i}$  is the soil matric suction in soil layer  $i$  ( $\text{m}$ );  $\theta_i$  is the water content in soil layer  $i$  ( $\text{m}^3 \text{ m}^{-3}$ );  $\theta_{33,i}$ , and  $\theta_{s,i}$  are the soil water content at field capacity and saturation, respectively ( $\text{m}^3 \text{ m}^{-3}$ ); and  $\lambda$  is the slope of the logarithmic suction-soil moisture curve (Saxton and Rawls, 2006).

#### 4.9 Gravity head

$$H_{g,i} = z_i$$

where  $H_{g,i}$  is the gravity head in soil layer  $i$  ( $\text{m}$ ); and  $z_i$  is the depth of the middle of soil layer  $i$  from the soil surface ( $\text{m}$ ).

#### 4.10 Hydraulic gradient

$$H_i = H_{m,i} + H_{g,i}$$

where in soil layer  $i$ ,  $H_i$  is the hydraulic gradient ( $\text{m}$ ),  $H_{m,i}$  is matric suction head ( $\text{m}$ ), and  $H_{g,i}$  is the gravity head ( $\text{m}$ ).

#### 4.11 Water content

**Water flux**

Darcy's law is used to describe the water flow in the soil. Water flow is taken to occur from the middle of layer  $i-1$  to the middle of layer  $i$ . The method based on Campbell (1994) is followed. Water flux into soil layer  $i$  is:

$$q_i = \begin{cases} \text{MIN}(K_{s,1}, P_{net,d}) - E_a - T_{a,i} & i = 1 \\ \bar{K}_{\theta,i} \frac{(H_i - H_{i-1})}{(z_i - z_{i-1})} - T_{ai} & 1 < i \leq N \\ K_{\theta,N} & i = N + 1 \end{cases}$$

$$\bar{K}_{\theta,i} = \frac{K_{\theta,i-1} - K_{\theta,i}}{\ln(K_{\theta,i-1}) - \ln(K_{\theta,i})}$$

where subscripts  $i-1$  and  $i$  denote soil layer  $i-1$  and  $i$ , respectively;  $q$  is the water flux ( $\text{m day}^{-1}$ );  $P_{net,d}$  is the net daily rainfall ( $\text{m day}^{-1}$ ; *discussed later*);  $E_a$  is the actual daily soil evaporation (occurs only from the first soil layer) ( $\text{m day}^{-1}$ );  $T_a$  is the daily extraction of water by roots (actual plant transpiration) ( $\text{m day}^{-1}$ );  $\bar{K}$  is the logarithmic mean of the hydraulic conductivities of layer  $i$  and  $i-1$  ( $\text{m day}^{-1}$ );  $K_{\theta}$  and  $s$  are the soil layer's hydraulic conductivity and thickness, respectively (m); and  $H$  is the hydraulic gradient (m). Note that for the first soil layer, water entry is via net rainfall,  $P_{net,d}$  whose entry into the soil is limited by the first soil layer's saturated hydraulic conductivity  $K_{s,1}$ .

*Note:* Water flux out of the last soil layer ( $i = N$ ) is denoted by  $q_{N+1}$ , and it is merely equal to  $K_{\theta,N}$  because of the assumption that the soil below the last layer is uniformly wet and it has the same water content as the last soil layer. Consequently, water flux is only due to gravity gradient (no matric suction gradient). In this case,  $q_{N+1} = K_{\theta,N}$ .

Also note that both  $E_a$  and  $T_{a,i}$  must be expressed in unit  $\text{m day}^{-1}$ .

### **Net flux**

$$\hat{q}_i = q_i - q_{i+1}$$

where  $\hat{q}_i$  is the net flux in soil layer  $i$  ( $\text{m day}^{-1}$ ); and  $q_i$  and  $q_{i+1}$  are the water flux into soil layer  $i$  and  $i+1$ , respectively ( $\text{m day}^{-1}$ ).

### **Water content**

$$\Theta_{i,t+1} = \Theta_{i,t} + \hat{q}_i$$

where  $\Theta_{i,t}$  and  $\Theta_{i,t+1}$  are the water depth in soil layer  $i$  (m) at time step  $t$  and  $t+1$ , respectively; and  $\hat{q}_i$  is the net flux in soil layer  $i$  ( $\text{m day}^{-1}$ ).

### **Water table (ground water)**

The water table, if specified, is assumed to have a constant water table level or depth and is located just beneath the last soil layer. For instance, if three soil layers are specified, with a

total soil layer thickness of 2 m, then the water table is located at a constant depth of 2 m from the soil surface. This water table is located just beneath the third (which is the last) soil layer.

Water flow from the water table follows Darcy's law as described earlier, where water flows upward, against the gravity, from the saturated water table into the drier soil layers above.

#### 4.12 Net rainfall

Net rainfall refers to the amount of rain reaching the ground as both throughfall and stemflow. It is determined by

$$\frac{P_{net,d}}{P_{g,d}} = MAX(0.7295, 1 - 0.0541L)$$

where  $P_{net,d}$  is the daily net rainfall (m);  $P_{g,d}$  is the daily gross rainfall (rainfall above the canopies, m); and  $L$  is the leaf area index ( $m^2$  leaf  $m^{-2}$  ground). The relationship between  $P_{net,d} / P_{g,d}$  and  $L$  is determined from Farmanta (2014).

Note:  $P_{net,d}$  must be in unit m before it is used in the soil water balance model.

#### 4.13 Daily integration

Hydraulic conductivity is sensitive to soil water content especially at near saturation. Consequently, it is vital that the water fluxes are calculated using very small time steps because using too large a time step (e.g., 1 day) may cause unrealistically large fluctuations or changes in the soil water content.

To overcome this problem, a specified number of iterations ( $f$ ) within a single time step is performed to simulate the daily water content. This method is the same as done by the Runge-Kutta numerical integration method.

*Note:* The larger the  $f$  value (i.e., the more iterations within a single time step), the more accurate the simulation results, but the slower the simulation run. The value of  $f$  depends on the hydraulic conductivity value,  $k$ ; a large  $k$  often means a large  $f$  is required for accurate results.

## CHAPTER 5. MAINTENANCE RESPIRATION

Food produced via photosynthesis is used for plant maintenance (supporting processes for continual plant survival) and growth (synthesis of new cells). The method by van Kraalingen *et al.* (1989) for oil palm is adapted.

### 5.1 Pinnae

$$M_{pinnae} = W_{pinnae} \times M_{c,pinnae} \times (24 - DL) / 24$$

where  $M_{pinnae}$  is the maintenance requirement for pinnae (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $W_{pinnae}$  is the dry weight of the pinnae (kg DM palm<sup>-1</sup>);  $DL$  is the day length (hour); and  $M_{c,pinnae}$  is the maintenance coefficient for the pinnae (kg CH<sub>2</sub>O kg<sup>-1</sup> DM), determined by

$$M_{c,pinnae} = (N_{pinnae} \times 0.036 \times 6.25) + (X_{pinnae} \times 0.072 \times X_c)$$

where  $N_{pinnae}$  and  $X_{pinnae}$  are the fractions by weight of nitrogen and mineral content in the pinnae, respectively (fraction); and  $X_c$  is the correction factor for mineral content (unitless), taken as 2.

### 5.2 Rachis

$$M_{rachis} = W_{rachis} \times M_{c,rachis}$$

where  $M_{rachis}$  is the maintenance requirement for the rachis (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $W_{rachis}$  is the dry weight of the rachis (kg DM palm<sup>-1</sup>); and  $M_{c,rachis}$  is the maintenance coefficient for the pinnae (kg CH<sub>2</sub>O kg<sup>-1</sup> DM), determined by

$$M_{c,rachis} = (N_{rachis} \times 0.036 \times 6.25) + (X_{rachis} \times 0.072 \times X_c)$$

where  $N_{rachis}$  and  $X_{rachis}$  are the fractions by weight of nitrogen and mineral content in the rachis, respectively (fraction); and  $X_c$  is the correction factor for mineral content (unitless), taken as 2.

### 5.3 Trunk

$$M_{trunk} = (W_{top,trunk} \times M_{c,trunk}) + (W_{bottom,trunk} \times M_{c,trunk} \times 0.06)$$

where  $M_{trunk}$  is the maintenance requirement for the trunk (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $W_{top,trunk}$  and  $W_{bottom,trunk}$  are the dry weight of the upper and bottom trunk, respectively (kg DM palm<sup>-1</sup>); and  $M_{c,trunk}$  is the maintenance coefficient for the trunk (kg CH<sub>2</sub>O kg<sup>-1</sup> DM), determined by

$$M_{c,trunk} = (N_{trunk} \times 0.036 \times 6.25) + (X_{trunk} \times 0.072 \times X_c)$$

where  $N_{trunk}$  and  $X_{trunk}$  are the fractions by weight of nitrogen and mineral content in the trunk, respectively (fraction); and  $X_c$  is the correction factor for mineral content (unitless), taken as 2. Both  $W_{top,trunk}$  and  $W_{bottom,trunk}$  are determined by

$$W_{top,trunk} = MIN(W_{trunk}, 45.0)$$

$$W_{bottom,trunk} = W_{trunk} - W_{top,trunk}$$

where  $W_{trunk}$  is the total dry weight of the trunk (kg DM palm<sup>-1</sup>).

#### 5.4 Roots

$$M_{roots} = W_{roots} \times M_{c,roots}$$

where  $M_{roots}$  is the maintenance requirement for the roots (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $W_{roots}$  is the dry weight of the roots (kg DM palm<sup>-1</sup>); and  $M_{c,roots}$  is the maintenance coefficient for the roots (kg CH<sub>2</sub>O kg<sup>-1</sup> DM), determined by

$$M_{c,roots} = (N_{roots} \times 0.036 \times 6.25) + (X_{roots} \times 0.072 \times X_c)$$

where  $N_{roots}$  and  $X_{roots}$  are the fractions by weight of nitrogen and mineral content in the roots, respectively (fraction); and  $X_c$  is the correction factor for mineral content (unitless), taken as 2.

#### 5.5 Generative organs

$$M_{organs} = (0.0027 \times W_{matbunch}) + (M_{c,rachis} \times W_{immbunch})$$

$$+ (M_{c,rachis} \times W_{maleflo})$$

where  $M_{organs}$  is the maintenance requirement for the generative organs (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $W_{matbunch}$ ,  $W_{immbunch}$ , and  $W_{maleflo}$  are the dry weights of the mature bunches, immature bunches, and male flowers, respectively (kg DM palm<sup>-1</sup>); and  $M_{c,rachis}$  is the maintenance coefficient for the rachis (kg CH<sub>2</sub>O kg<sup>-1</sup> DM).

#### 5.6 Total maintenance

$$M'_{total} = M_{pinnae} + M_{rachis} + M_{trunk} + M_{roots} + M_{organs} + M_{metabolic}$$

where  $M'_{total}$  is the total maintenance requirement for the whole tree, which is the summation of the maintenance requirement for pinnae ( $M_{pinnae}$ ), rachis ( $M_{rachis}$ ), trunk ( $M_{trunk}$ ), roots ( $M_{roots}$ ), generative organs ( $M_{organs}$ ), as well as metabolic activity ( $M_{metabolic}$ ) (all units in kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>).  $M_{metabolic}$  is determined by

$$M_{metabolic} = 0.16 \Lambda_{canopy,d} / W_{total}$$

where  $\Lambda_{canopy,d}$  is the daily canopy photosynthesis (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>); and  $W_{total}$  is the total dry weight of all plant parts:

$$W_{total} = W_{pinnae} + W_{rachis} + W_{trunk} + W_{roots} + W_{organs}$$

where the weights for pinnae ( $W_{pinnae}$ ), rachis ( $W_{rachis}$ ), trunk ( $W_{trunk}$ ), roots ( $W_{roots}$ ), generative organs ( $W_{organs}$ ) are summed (all weights in kg DM palm<sup>-1</sup>).

Note that maintenance respiration is not adjusted or corrected for air temperature. However, assimilates can all be diverted for maintenance respiration if the air temperature is lower than 15 °C or higher than 45 °C:

$$M_{total} = \begin{cases} M'_{total} \times T_c & 15^\circ C < T_{mean} < 45^\circ C \\ \Lambda_{canopy,d} & \text{otherwise} \end{cases}$$

where  $M_{total}$  is the corrected maintenance respiration for temperature (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $T_{mean}$  is the daily average of the minimum ( $T_{min}$ ) and maximum ( $T_{max}$ ) air temperature (°C). Consequently, this means that oil palm only grows with air temperatures between 15 to 45 °C.

The amount of assimilates available for growth respiration is the leftover assimilates after maintenance respiration, or

$$G_{growth} = MAX(0, \Lambda_{canopy,d} - M_{total})$$

where  $G_{growth}$  is the amount of assimilates available for growth respiration (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>).

## CHAPTER 6. GROWTH RESPIRATION

### 6.1 Annual VDM requirement

Annual vegetative dry matter (VDM) requirement is determined based on leaf area index, following a rectangular hyperbola relationship, as

$$VDM_{yr} = \left( a + \frac{b}{L^{1.5}} \right)^{-1}$$

where  $VDM_{yr}$  is the annual VDM requirement (kg DM palm<sup>-1</sup> year<sup>-1</sup>);  $L$  is the leaf area index (m<sup>2</sup> leaf m<sup>-2</sup> ground); and coefficients  $a$  and  $b$  are determined by

$$a = \frac{A}{VDM_{max,PD}}$$

$$b = 0.1 \left( \frac{1}{A} - 1 \right) \left( \frac{PD}{100} \right)^{\frac{1}{A}}$$

$$VDM_{max,PD} = 231 PD^{1-\frac{1}{A}}$$

$$A = 0.935$$

where  $PD$  is the planting density (palms ha<sup>-1</sup>); and for the given planting density,  $VDM_{max,PD}$  is the maximum VDM (kg DM palm<sup>-1</sup> year<sup>-1</sup>) and  $L_{max,PD}$  is the maximum leaf area index (m<sup>2</sup> leaf m<sup>-2</sup> ground) for a given  $PD$  planting density.

Coefficients,  $a$ ,  $b$ , and  $A$ , as well as  $VDM_{max,PD}$ , were obtained from fitting the best functions to measured data from van Kraalingen *et al.* (1989).

### 6.2 Daily VDM requirement

$$VDM_d = \frac{VDM_{yr}}{365}$$

where  $VDM_{yr}$  and  $VDM_d$  are the annual and daily VDM requirement, respectively (kg DM palm<sup>-1</sup> day<sup>-1</sup>).

### 6.3 Dry matter partitioning

$DM_{pinnae}$ ,  $DM_{rachis}$ ,  $DM_{trunk}$ , and  $DM_{roots}$  are the fractions of dry matter partitioned to pinnae, rachis, trunk, and roots, respectively (fraction). Their constant values are as follows:

$$DM_{pinnae} = 0.24$$



$$DM_{rachis} = 0.46$$

$$DM_{trunk} = 0.14$$

$$DM_{roots} = 1 - (DM_{pinnae} + DM_{rachis} + DM_{trunk})$$

where *age* is the tree age (days). Dry matter partitioning for pinnae, rachis, and trunk are mean values of measured dry matter partitioning from Corley *et al.* (1971), Henson (1995), and Henson and Mohd Tayeb (2003).

#### 6.4 Growth rates

$G_{pinnae}$ ,  $G_{rachis}$ ,  $G_{trunk}$ , and  $G_{roots}$  are the growth rates for pinnae, rachis, trunk, and roots, respectively (kg DM palm<sup>-1</sup> day<sup>-1</sup>). They are determined by

$$G_{pinnae} = DM_{pinnae} \times A \times CVF$$

$$G_{rachis} = DM_{rachis} \times A \times CVF$$

$$G_{trunk} = DM_{trunk} \times A \times CVF$$

$$G_{roots} = DM_{roots} \times A \times CVF$$

where  $A$  (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>) is:

$$A = \min\left(\frac{VDM_d}{CVF}, G_{growth}\right)$$

and  $CVF$  is a factor (kg DM kg<sup>-1</sup> CH<sub>2</sub>O) to convert a weight expressed on a CH<sub>2</sub>O weight basis to that on a dry matter (DM) basis.  $CVF$  is determined by

$$CVF = (0.70 \times DM_{pinnae}) + (0.70 \times DM_{rachis}) \\ + (0.66 \times DM_{trunk}) + (0.65 \times DM_{roots})$$

where  $DM_{pinnae}$ ,  $DM_{rachis}$ ,  $DM_{trunk}$ , and  $DM_{roots}$  are the fractions of dry matter partitioned to pinnae, rachis, trunk, and roots, respectively (fraction) (van Kraalingen *et al.*, 1989). In the above equation,  $VDM_d$  is the daily VDM requirement (kg DM palm<sup>-1</sup> day<sup>-1</sup>) and  $G_{growth}$  is the amount of assimilates available for growth respiration (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>).

#### 6.5 Death rates of pinnae and rachis

Leaves (pinnae and rachis) deaths are caused by aging, and they are determined by

$$G_{death,leaves} = \begin{cases} 0 & age \leq 600 \\ 0.0016 \left( \frac{age - 600}{2500 - 600} \right) & 600 < age \leq 2500 \\ 0.0016 & age > 2500 \end{cases}$$

where  $G_{death,leaves}$  is the death rate of both pinnae and rachis due to natural causes (kg DM palm<sup>-1</sup> day<sup>-1</sup>); and  $age$  is the tree age (days). Values for the death rate of leaves were obtained from Dr. Ian E. Henson, then at Malaysian Palm Oil Board (personal communication).

## 6.6 Death rates of roots

$$G_{death,roots} = \begin{cases} 0 & age \leq 1200 \\ \frac{0.00009592age - 0.11510791}{365} & 1200 < age \leq 3285 \\ \frac{0.2}{365} & age > 3285 \end{cases}$$

where  $G_{death,roots}$  is the death rate of roots due to natural causes (kg DM palm<sup>-1</sup> day<sup>-1</sup>); and  $age$  is the tree age (days) (based on unpublished data).

## 6.7 Assimilates available for generative growth

The amount of assimilates available for generative growth is the leftover assimilates after growth respiration, or

$$G_{gen} = MAX \left( 0, G_{growth} - \frac{VDM_d}{CVF} \right)$$

where  $G_{gen}$  is the amount of assimilates available for generative organs growth (kg CH<sub>2</sub>O palm<sup>-1</sup> day<sup>-1</sup>);  $VDM_d$  is daily VDM requirement (kg DM palm<sup>-1</sup> day<sup>-1</sup>); and  $CVF$  is a factor (kg DM kg<sup>-1</sup> CH<sub>2</sub>O) to convert dry matter weight into CH<sub>2</sub>O weight.

## 6.8 Plant part weights

The plant part weights are incremented using Euler's integration method:

$$W_{t2} = W_{t1} + (rate_{t1} \times \Delta t)$$

where  $W_{t1}$  and  $W_{t2}$  are the weights at time  $t1$  and  $t2$ , respectively;  $rate_{t1}$  is the rate of increase at time  $t1$ ; and  $\Delta t$  is the time interval, or  $(t2 - t1)$ . Since the model time step is always set at one day,  $\Delta t = 1$ . Thus,  $\Delta t$  can be ignored in subsequent calculations.

The various plant part weights are updated as follows:

$$W_{pinnae,t2} = W_{pinnae,t1} + (G_{pinnae} - G_{death,leaves})$$

$$W_{rachis,t2} = W_{rachis,t1} + (G_{rachis} - G_{death,leaves})$$

$$W_{trunk,t2} = W_{trunk,t1} + G_{trunk}$$

$$W_{roots,t2} = W_{roots,t1} + (G_{roots} - G_{death,roots})$$

where  $W_{pinnae}$ ,  $W_{rachis}$ ,  $W_{trunk}$ , and  $W_{roots}$  are the dry weights for pinnae, rachis, trunk, and roots, respectively (kg DM palm<sup>-1</sup>);  $G_{pinnae}$ ,  $G_{rachis}$ ,  $G_{trunk}$ , and  $G_{roots}$  are the growth rates for pinnae, rachis, trunk, and roots, respectively (kg DM palm<sup>-1</sup> day<sup>-1</sup>);  $G_{death,leaves}$  is the death rate of both pinnae and rachis due to natural causes (kg DM palm<sup>-1</sup> day<sup>-1</sup>); and  $G_{death,roots}$  is the death rate of roots due to aging (kg DM palm<sup>-1</sup> day<sup>-1</sup>) (van Kraalingen *et al.*, 1989).

## 6.9 Leaf area index

Leaf area index  $LAI$  (m<sup>2</sup> leaf m<sup>-2</sup> ground) is determined by:

$$LAI = W_{pinnae} \times SLA \times PD / 10000$$

where  $SLA$  is the specific leaf area (m<sup>2</sup> leaf kg<sup>-1</sup> DM);  $W_{pinnae}$  is the weight of the pinnae (kg DM palm<sup>-1</sup>); and  $PD$  is the planting density (palms ha<sup>-1</sup>).  $SLA$  was measured for several palm ages, and the following relationship was obtained:

$$SLA = \begin{cases} 13.2 & age \leq 365 \\ 7.25 & age \geq 840 \end{cases}$$

where  $age$  is the palm age (in days). For palm ages between 365 and 840 days,  $SLA$  is obtained by linear interpolation between values 13.2 and 7.25 m<sup>2</sup> leaf kg<sup>-1</sup> DM.



The algorithm of this boxcar train technique essentially consists of two steps: 1) the weight of a generative organ is shifted to successively older age classes, and 2) the weight is incremented according to the calculated growth rate. The lifespan of male flowers is set at 240 days, after which they will die. For immature bunches, however, the day after 240 days marks the point of pollination, after which the growth of the pollinated bunch (*i.e.*, mature bunch) will occur.

To determine the growth rates of the generative organs, the following equations are used:

$$G_{immbunch} = \frac{(f_{e,immbunch} \times G_{gen} \times CVF_2)}{C}$$

$$G_{matbunch} = \frac{(f_{e,matbunch} \times G_{gen} \times CVF_2)}{D}$$

$$G_{maleflo} = \frac{(f_{e,maleflo} \times G_{gen} \times CVF_2)}{E}$$

where  $G_{immbunch}$ ,  $G_{matbunch}$ , and  $G_{maleflo}$  are the growth rates for immature bunches, mature bunches, and male flowers, respectively ( $\text{kg DM day}^{-1}$ );  $G_{gen}$  is the amount of assimilates available for generative growth, after accounting for maintenance and growth respiration ( $\text{kg DM day}^{-1}$ );  $CVF_2$  is a factor ( $\text{kg DM kg}^{-1} \text{ CH}_2\text{O}$ ) to convert dry matter weight into  $\text{CH}_2\text{O}$  weight, determined by

$$CVF_2 = (0.70 \times f_{e,immbunch}) + (0.44 \times f_{e,matbunch}) + (0.70 \times f_{e,maleflo})$$

where  $f_{e,immbunch}$ ,  $f_{e,matbunch}$ , and  $f_{e,maleflo}$  are the scaled fraction of dry matter to immature bunches, mature bunches, and male flowers, respectively (fraction), and they are determined by

$$f_{e,immbunch} = f_{immbunch} / f_{total}$$

$$f_{e,matbunch} = f_{matbunch} / f_{total}$$

$$f_{e,maleflo} = f_{maleflo} / f_{total}$$

where

$$f_{total} = f_{immbunch} + f_{matbunch} + f_{maleflo}$$

$$f_{immbunch} = \frac{DM_{immbunch} \times C}{240}$$

$$f_{matbunch} = \frac{DM_{matbunch} \times D}{180}$$

$$f_{maleflo} = \frac{DM_{maleflo} \times E}{240}$$

where  $DM_{immbunch}$ ,  $DM_{matbunch}$ , and  $DM_{maleflo}$  are the fractions of available assimilates for generative growth partitioned to immature bunches, mature bunches, and male flowers,

respectively (fraction); and  $C$ ,  $D$ , and  $E$  are the numbers of non-zero weights for immature bunches, mature bunches, and male flowers, respectively (van Kraalingen *et al.*, 1989).

*Note:*  $DM_{imbunch}$ ,  $DM_{matbunch}$ , and  $DM_{maleflo}$  are taken as constants 0.159, 0.682, and 0.159, respectively, and denominator 240 is the growing time for both immature bunches and male flowers, whereas 180 is for mature bunches (van Kraalingen *et al.*, 1989).

Also note that plant storage is not modelled.

## 7.2 Organ weights and yield

The generative organ weights are incremented simply by

$$w_{1,i}(t2) = \begin{cases} 0 & w_{1,i}(t1) = 0 \\ w_{1,i}(t1) + G_{maleflo} & w_{1,i}(t1) > 0 \end{cases}$$

$$w_{2,j}(t2) = \begin{cases} 0 & w_{2,j}(t1) = 0 \\ w_{2,j}(t1) + G_{imbunch} & w_{2,j}(t1) > 0 \end{cases}$$

$$w_{3,k}(t2) = \begin{cases} 0 & w_{3,k}(t1) = 0 \\ w_{3,k}(t1) + G_{matbunch} & w_{3,k}(t1) > 0 \end{cases}$$

where  $w_{1,i}$  is the dry weight of male flowers in age class  $i$  (kg DM);  $w_{2,j}$  is the dry weight of immature bunches in age class  $j$  (kg DM);  $w_{3,k}$  is the dry weight of mature bunches in age class  $k$  (kg DM); and  $G_{imbunch}$ ,  $G_{matbunch}$ , and  $G_{maleflo}$  are the growth rates for immature bunches, mature bunches, and male flowers, respectively (kg DM day<sup>-1</sup>). The organ weights are only incremented for non-zero preceding weights (van Kraalingen *et al.*, 1989).

The total weight for a given generative organ is the summation of weights in all the age classes, or

$$W_{maleflo} = \sum_{i=1}^{240} w_{1,i}$$

$$W_{imbunch} = \sum_{j=1}^{240} w_{2,j}$$

$$W_{matbunch} = \sum_{k=1}^{180} w_{3,k}$$

where  $W_{matbunch}$ ,  $W_{imbunch}$ , and  $W_{maleflo}$  are the dry weights of the mature bunches, immature bunches, and male flowers, respectively (kg DM palm<sup>-1</sup>);  $w_{1,i}$  is the dry weight of male flowers in age class  $i$  (kg DM);  $w_{2,j}$  is the dry weight of immature bunches in age class  $j$  (kg DM); and  $w_{3,k}$  is the dry weight of mature bunches in age class  $k$  (kg DM) (van Kraalingen *et al.*, 1989).

*Note:* Oil palm yield is  $w_{3,k+1}$ .

## CHAPTER 8. MISCELLANEOUS

### 8.1 Trunk, canopy, and tree height

The initial trunk height is a function of both tree age and planting density. It can be described by

$$h_{trunk,(t0)} = \exp\left(a + \frac{b}{PD^2} + \frac{c}{age}\right)$$

$$a = 2.845586$$

$$b = -1980.88805$$

$$c = -5166.36569$$

where  $h_{trunk,(t0)}$  is the initial trunk height (m);  $PD$  is the planting density (palms ha<sup>-1</sup>); and  $age$  is the tree age (days). Coefficients,  $a$ ,  $b$ , and  $c$ , were calculated based on oil palm trunk height data from Jacquemard (1979, 1998), Breure and Powell (1988), Rao *et al.* (1992), Kwan (1994), and Henson and Mohd Tayeb (2003).

The rate of trunk height increase is described by

$$h'_{trunk} = -\frac{c}{0.7age^2} \times \exp\left(a + \frac{b}{PD^2} + \frac{c}{age}\right) \times (0.21f_{water} + 0.553)$$

where  $h'_{trunk}$  is the rate of trunk height increase (m day<sup>-1</sup>);  $PD$  is the planting density (palms ha<sup>-1</sup>); and  $age$  is the tree age (days);  $f_{water}$  is the water stress function, determined as the ratio between actual and potential transpiration (0 to 1); and the values for the coefficients  $a$ ,  $b$ , and  $c$  are as defined previously. Consequently, trunk height is updated at every model step of 1 day as

$$h_{trunk,(t2)} = h_{trunk,(t1)} + h'_{trunk}$$

The canopy height  $h_{canopy}$  (m) is determined by

$$h_{canopy} = 1.5091 + 0.001382age$$

where  $age$  is the tree age (days). Consequently, the full tree height  $h$  (m) is determined by

$$h = h_{trunk} + h_{canopy}$$

### 8.2 Rooting depth

The rooting depth is determined by

$$d_{root,(t2)} = \text{MIN}\left[S_N, d_{root,(t1)} + (dg_{root} \times f_{water})\right]$$



where  $d_{root}$  is the rooting depth (m);  $dg_{root}$  is the rate of increase in the rooting depth ( $\text{m day}^{-1}$ );  $f_{water}$  is the water stress function, determined as the ratio between actual and potential transpiration (0 to 1); and  $S_N$  is the total thickness of the soil profile (m). In other words, the increase in rooting depth is adversely effected by water stress, and rooting depth will not exceed the total thickness of the soil profile being studied.

*Note:*  $dg_{root}$  is taken as a constant at  $0.002 \text{ m day}^{-1}$ .

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## APPENDIX A. LIST OF MAIN SYMBOLS

The following lists all the main symbols used in the model. Note that some symbols have more than one meaning, depending on the context.

| Symbol                                          | Description                                                                                | Unit                                                        |
|-------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| $\alpha$                                        | extinction coefficient for wind speed                                                      | -                                                           |
| $\alpha$                                        | fraction of PAR absorbed (0.8)                                                             | -                                                           |
| $\alpha$                                        | scattering coefficient (0.5 for total solar radiation, or 0.8 for PAR)                     | -                                                           |
| $\beta$                                         | solar elevation or height (from horizontal)                                                | radians                                                     |
| $\delta$                                        | solar declination                                                                          | radians                                                     |
| $\Delta$                                        | slope of the saturated vapor pressure curve                                                | mbar K <sup>-1</sup>                                        |
| $\phi$                                          | solar azimuth (angle clockwise from North)                                                 | radians                                                     |
| $\phi_p$                                        | soil porosity                                                                              | -                                                           |
| $\gamma$                                        | psychometric constant (0.658)                                                              | mbar K <sup>-1</sup>                                        |
| $\Gamma^*$                                      | CO <sub>2</sub> compensation point                                                         | μmol mol <sup>-1</sup>                                      |
| $\lambda$                                       | latent heat of vaporization of water (2454000)                                             | J kg <sup>-1</sup>                                          |
| $\lambda$                                       | pore-size distribution index                                                               | -                                                           |
| $\lambda$                                       | site latitude                                                                              | radians                                                     |
| $\lambda ET$                                    | total latent heat flux density                                                             | W m <sup>-2</sup>                                           |
| $\lambda ET_c$                                  | latent heat flux density of crop                                                           | W m <sup>-2</sup>                                           |
| $\lambda ET_s$                                  | latent heat flux density of soil                                                           | W m <sup>-2</sup>                                           |
| $\Lambda_{canopy}$                              | gross canopy assimilation rate of CO <sub>2</sub> for the given hour                       | μmol CO <sub>2</sub> m <sup>-2</sup> ground s <sup>-1</sup> |
| $\Lambda_{canopy,d}$ ,<br>$\Lambda'_{canopy,d}$ | daily gross photosynthesis with and without O <sub>3</sub> reduction effects, respectively | kg CH <sub>2</sub> O palm <sup>-1</sup> day <sup>-1</sup>   |
| $\Lambda_{sl/sh}$                               | gross leaf assimilation rate of CO <sub>2</sub> for sunlit (sl) or shaded (sh) leaves      | μmol CO <sub>2</sub> m <sup>-2</sup> leaf s <sup>-1</sup>   |
| $\pi$                                           | pi constant (3.1428571)                                                                    | -                                                           |
| $\theta$                                        | solar inclination (from vertical)                                                          | radians                                                     |

|                                             |                                                                                                   |                                            |
|---------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------|
| $\theta_i$                                  | volumetric water content for soil layer $i$                                                       | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_s$                                  | saturated volumetric water content                                                                | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_{33}$                               | volumetric water content at field capacity                                                        | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_{1500}$                             | volumetric water content at permanent wilting point                                               | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_{root}$                             | volumetric water content in the root zone                                                         | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_{cr,root}$                          | critical volumetric water content in the root zone                                                | $\text{m}^3 \text{m}^{-3}$                 |
| $\theta_{s,root}$ ,<br>$\theta_{1500,root}$ | volumetric water content at saturation and permanent wilting point in the root zone, respectively | $\text{m}^3 \text{m}^{-3}$                 |
| $\Theta_i$                                  | soil water content for soil layer $I$ , expressed as water depth                                  | m                                          |
| $\rho c_p$                                  | volumetric heat capacity of air (1221.09)                                                         | $\text{J m}^{-3} \text{K}^{-1}$            |
| $\sigma$                                    | Stefan-Boltzmann constant ( $5.67 \times 10^{-8}$ )                                               | $\text{W m}^{-2} \text{K}^{-4}$            |
| $\tau$                                      | hour angle                                                                                        | radians                                    |
| $\tau$                                      | CO <sub>2</sub> / O <sub>2</sub> specificity factor                                               | $\mu\text{mol } \mu\text{mol}^{-1}$        |
| $\tau$                                      | soil tortuosity                                                                                   | -                                          |
| $\tau$                                      | O <sub>3</sub> residence time                                                                     | s                                          |
| $\tau_{dr,\alpha}$                          | penetration function for direct solar radiation (corrected for scattering)                        | -                                          |
| $\omega$                                    | canopy clustering coefficient                                                                     | -                                          |
| $\omega_0$                                  | canopy clustering coefficient when the sun is at zenith                                           | -                                          |
| $\psi_e$                                    | air entry suction                                                                                 | m                                          |
| $age$                                       | tree age                                                                                          | days                                       |
| $A$                                         | total energy supply                                                                               | $\text{W m}^{-2}$                          |
| $A_c$                                       | fraction of energy available to the plant                                                         | $\text{W m}^{-2}$                          |
| $A_s$                                       | fraction of energy available to the soil                                                          | $\text{W m}^{-2}$                          |
| $C$                                         | fractional clay content in soil                                                                   | -                                          |
| $C_a$                                       | ambient CO <sub>2</sub> concentration in air                                                      | $\mu\text{mol mol}^{-1}$                   |
| $C_i$                                       | intercellular CO <sub>2</sub> concentration                                                       | $\mu\text{mol mol}^{-1}$                   |
| $CVF, CVF_2$                                | conversion from CH <sub>2</sub> O to DM weight                                                    | $\text{kg DM kg}^{-1} \text{CH}_2\text{O}$ |

|                               |                                                                                      |                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| $d$                           | zero plane displacement                                                              | m                                                                    |
| $d_{root}$                    | rooting depth                                                                        | m                                                                    |
| $dg_{root}$                   | rooting depth growth rate                                                            | m day <sup>-1</sup>                                                  |
| $D$                           | vapor pressure deficit                                                               | mbar                                                                 |
| $D_0$                         | vapor pressure deficit at the mean canopy flow                                       | mbar                                                                 |
| $D_{leaf}$                    | leaf vapour pressure deficit                                                         | mbar                                                                 |
| $D_{min}$                     | minimum vapour pressure deficit                                                      | mbar                                                                 |
| $D_{m,v}$                     | vapor diffusion coefficient in air (24.7 x 10 <sup>-6</sup> )                        | m <sup>2</sup> s <sup>-1</sup>                                       |
| $DL$                          | day length                                                                           | hour                                                                 |
| $DM_x$                        | fraction of DM to plant part $x$ , where $x$ is for pinnae, rachis, trunk, and roots | -                                                                    |
| $e_a$                         | air vapor pressure                                                                   | mbar                                                                 |
| $e_m$                         | quantum efficiency or quantum yield                                                  | μmol μmol <sup>-1</sup>                                              |
| $e_s \llbracket T \rrbracket$ | function returning the saturated vapor pressure at air temperature $T$               | mbar                                                                 |
| $E_a$                         | actual soil evaporation                                                              | m day <sup>-1</sup>                                                  |
| $E_{iso,sl/sh}$               | emitted isoprene concentration by sunlit (sl) or shaded (sh) leaves                  | g C <sub>5</sub> H <sub>8</sub> g <sup>-1</sup> leaf s <sup>-1</sup> |
| $ET_c$                        | potential transpiration                                                              | m day <sup>-1</sup>                                                  |
| $ET_s$                        | potential evaporation                                                                | m day <sup>-1</sup>                                                  |
| $f_D$                         | reduction to maximum stomatal conductance due to vapour pressure deficit             | -                                                                    |
| $f_{iso}$                     | CO <sub>2</sub> -inhibited isoprene emission function                                | -                                                                    |
| $f_{O3}$                      | fraction of assimilates reduced due to O <sub>3</sub> dose                           | -                                                                    |
| $f_{PAR}$                     | reduction to maximum stomatal conductance due to PAR                                 | -                                                                    |
| $f_{water}$                   | reduction to maximum stomatal conductance due to water stress                        | -                                                                    |
| $F_{O3eff}$                   | O <sub>3</sub> effective dose                                                        | mmol O <sub>3</sub> m <sup>-2</sup> ground                           |
| $F_{stom}^*$                  | instantaneous O <sub>3</sub> dose                                                    | μmol O <sub>3</sub> m <sup>-2</sup> ground s <sup>-1</sup>           |
| $gst$                         | stomatal conductance                                                                 | m s <sup>-1</sup>                                                    |



|                                                     |                                                                                   |                                                     |
|-----------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|
| $gst_{max}$                                         | maximum stomatal conductance                                                      | $\text{m s}^{-1}$                                   |
| $G$                                                 | soil (ground) heat flux density                                                   | $\text{W m}^{-2}$                                   |
| $G_{gen}$                                           | assimilates available to generative organs growth                                 | $\text{kg CH}_2\text{O palm}^{-1} \text{ day}^{-1}$ |
| $G_{growth}$                                        | assimilates available to growth respiration                                       | $\text{kg CH}_2\text{O palm}^{-1} \text{ day}^{-1}$ |
| $G_{maleflo},$<br>$G_{immbunch},$<br>$G_{matbunch}$ | growth rate of male flowers, immature bunches, and mature bunches, respectively   | $\text{kg DM day}^{-1}$                             |
| $G_x$                                               | growth rate of plant part $x$ , where $x$ is for pinnae, rachis, trunk, and roots | $\text{kg DM palm}^{-1} \text{ day}^{-1}$           |
| $G_{death,leaves}$                                  | death rate of both pinnae and rachis due to natural causes                        | $\text{kg DM palm}^{-1} \text{ day}^{-1}$           |
| $G_{death,roots}$                                   | death rate of roots due to natural causes                                         | $\text{kg DM palm}^{-1} \text{ day}^{-1}$           |
| $h$                                                 | full tree height (trunk + canopy)                                                 | $\text{m}$                                          |
| $h_{canopy}$                                        | canopy height                                                                     | $\text{m}$                                          |
| $h_{trunk}$                                         | trunk height                                                                      | $\text{m}$                                          |
| $h'_{trunk}$                                        | rate of growth for trunk height                                                   | $\text{m day}^{-1}$                                 |
| $H_i$                                               | soil hydraulic gradient for soil layer $i$                                        | $\text{m}$                                          |
| $H_c$                                               | sensible heat flux density of crop                                                | $\text{W m}^{-2}$                                   |
| $H_{g,i}$                                           | gravity head for soil layer $i$                                                   | $\text{m}$                                          |
| $H_{m,i}$                                           | matric suction head for soil layer $i$                                            | $\text{m}$                                          |
| $H_s$                                               | sensible heat flux density of soil                                                | $\text{W m}^{-2}$                                   |
| $I_c$                                               | solar constant                                                                    | $\text{W m}^{-2}$                                   |
| $I_{df}$                                            | diffuse solar irradiance                                                          | $\text{W m}^{-2}$                                   |
| $I_{dr}$                                            | direct solar irradiance                                                           | $\text{W m}^{-2}$                                   |
| $I_{et}$                                            | extraterrestrial solar irradiance                                                 | $\text{W m}^{-2}$                                   |
| $I_{sl/sh}$                                         | total solar irradiance on sunlit (sl) or shaded (sh) leaves                       | $\text{W m}^{-2}$                                   |
| $I_t$                                               | total solar irradiance                                                            | $\text{W m}^{-2}$                                   |
| $I_{df,d}$                                          | daily diffuse solar irradiance                                                    | $\text{J m}^{-2} \text{ day}^{-1}$                  |
| $I_{dr,d}$                                          | daily direct solar irradiance                                                     | $\text{J m}^{-2} \text{ day}^{-1}$                  |
| $I_{et,d}$                                          | daily extraterrestrial solar irradiance                                           | $\text{J m}^{-2} \text{ day}^{-1}$                  |
| $I_{t,d}$                                           | daily total solar irradiance                                                      | $\text{J m}^{-2} \text{ day}^{-1}$                  |

|                                                             |                                                                                                                          |                                                           |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| $k$                                                         | von Karman's constant (0.4)                                                                                              | -                                                         |
| $k_{df}$                                                    | canopy extinction coefficient for diffuse solar radiation (corrected for canopy clustering)                              | -                                                         |
| $k_{dr}$                                                    | canopy extinction coefficient for direct radiation                                                                       | -                                                         |
| $K_{\theta,i}$                                              | soil hydraulic conductivity for soil layer $i$                                                                           | m day <sup>-1</sup>                                       |
| $\bar{K}_{\theta,i}$                                        | logarithmic mean of hydraulic conductivity for soil layer $i$ and $(i-1)$                                                | m day <sup>-1</sup>                                       |
| $K_c$                                                       | Michaelis-Menten constant for CO <sub>2</sub>                                                                            | μmol mol <sup>-1</sup>                                    |
| $K_o$                                                       | Michaelis-Menten constant for O <sub>2</sub>                                                                             | μmol mol <sup>-1</sup>                                    |
| $K_{s,i}$                                                   | saturated hydraulic conductivity for soil layer $i$                                                                      | m day <sup>-1</sup>                                       |
| $K_T$                                                       | sky clearness index                                                                                                      | -                                                         |
| $l$                                                         | dry soil layer thickness                                                                                                 | m                                                         |
| $l$                                                         | air mixing height                                                                                                        | m                                                         |
| $l$                                                         | mean pinnae length                                                                                                       | m                                                         |
| $l_m$                                                       | mean distance between pinnae                                                                                             | m                                                         |
| $L$                                                         | leaf area index                                                                                                          | m <sup>2</sup> leaf m <sup>-2</sup> ground                |
| $L_{cr}$                                                    | critical leaf area index                                                                                                 | m <sup>2</sup> leaf m <sup>-2</sup> ground                |
| $L_{sh}$                                                    | shaded leaf area index                                                                                                   | m <sup>2</sup> leaf m <sup>-2</sup> ground                |
| $L_{sl}$                                                    | sunlit leaf area index                                                                                                   | m <sup>2</sup> leaf m <sup>-2</sup> ground                |
| $L_{maxPD}$                                                 | maximum leaf area index for a given $PD$ planting density (palms ha <sup>-1</sup> )                                      | m <sup>2</sup> leaf m <sup>-2</sup> ground                |
| $M_{total}, M'_{total}$                                     | total maintenance respiration requirement with and without temperature correction, respectively                          | kg CH <sub>2</sub> O palm <sup>-1</sup> day <sup>-1</sup> |
| $M_x$                                                       | maintenance requirement for plant part $x$ , where $x$ is for pinnae, rachis, trunk, roots, organs, metabolic, and total | kg CH <sub>2</sub> O palm <sup>-1</sup> day <sup>-1</sup> |
| $M_{c,x}$                                                   | maintenance coefficient for plant part $x$ , where $x$ is for pinnae, rachis, trunk, and roots                           | kg CH <sub>2</sub> O kg <sup>-1</sup> DM                  |
| $MAX(x_1, x_2, \dots, x_n),$<br>$MIN(x_1, x_2, \dots, x_n)$ | functions to return the maximum and minimum values of $x_1, x_2, \dots, x_n$ , respectively                              | -                                                         |
| $n$                                                         | extinction coefficient for eddy diffusivity                                                                              | -                                                         |

|                              |                                                                                                               |                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| $N_x$                        | fraction by weight of nitrogen content in plant part $x$ , where $x$ is for pinnae, rachis, trunk, and roots. | -                                                       |
| $O_a$                        | ambient concentration of $O_2$ in air                                                                         | $\mu\text{mol mol}^{-1}$                                |
| $O_z$                        | ground level $O_3$ concentration for the given hour                                                           | $\text{nmol mol}^{-1}$                                  |
| $O_{z,d}$                    | ground level $O_3$ concentration for the given day of year                                                    | $\text{nmol mol}^{-1}$                                  |
| $O_{z,loss}$                 | fractional rate of $O_3$ chemical reaction in air                                                             | $\text{s}^{-1}$                                         |
| $O_{z,net}$                  | total $O_3$ concentration (concentrations of both background $O_3$ and $O_3$ formed from isoprene emission)   | $\mu\text{g } O_3 \text{ m}^{-2} \text{ ground s}^{-1}$ |
| $O_{z,prod}$                 | total $O_3$ concentration created from isoprene emitted by sunlit and shaded leaves                           | $\text{g } C_5H_8 \text{ g}^{-1} \text{ leaf s}^{-1}$   |
| $O_{z,sl/sh}$                | $O_3$ concentration created from isoprene emitted by sunlit (sl) or shaded (sh) leaves                        | $\text{g } O_3 \text{ palm}^{-1} \text{ s}^{-1}$        |
| $\overline{O_{z,yr}}$        | average ground level $O_3$ concentration for the year                                                         | $\text{nmol mol}^{-1}$                                  |
| $O_{z,yr,max}, O_{z,yr,min}$ | maximum and minimum ground level $O_3$ concentration for the year                                             | $\text{nmol mol}^{-1}$                                  |
| $OM$                         | organic matter content in soil                                                                                | %                                                       |
| $p$                          | surface albedo (reflection)                                                                                   | -                                                       |
| $P_a$                        | atmospheric pressure (101)                                                                                    | kPa                                                     |
| $PD$                         | planting density                                                                                              | $\text{palms ha}^{-1}$                                  |
| $q_i$                        | water flux into soil layer $i$                                                                                | $\text{m day}^{-1}$                                     |
| $\hat{q}_i$                  | net water flux into soil layer $i$                                                                            | $\text{m day}^{-1}$                                     |
| $Q_{df}$                     | diffuse PAR component (above canopies)                                                                        | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$              |
| $Q_{dr}$                     | direct PAR component (above canopies)                                                                         | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$              |
| $Q_{sh}$                     | total PAR absorbed by shaded leaves                                                                           | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$              |
| $Q_{sl}$                     | total PAR absorbed by sunlit leaves                                                                           | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$              |
| $Q_{10,\xi}$                 | relative change in a parameter $\xi$ for every 10 °C change                                                   | -                                                       |

|                       |                                                                                        |                                              |
|-----------------------|----------------------------------------------------------------------------------------|----------------------------------------------|
| $\bar{Q}_{p,df}$      | mean diffuse component of PAR (within canopies)                                        | $\mu\text{mol m}^{-2} \text{ ground s}^{-1}$ |
| $Q_{p,dr}$            | unintercepted direct component of PAR (with scattering component) (within canopies)    | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$   |
| $Q_{p,dr,\alpha}$     | PAR scattered component only (within canopies)                                         | $\mu\text{mol m}^{-2} \text{ ground s}^{-1}$ |
| $Q_{p,dr,dr}$         | unintercepted direct component of PAR (without scattering component) (within canopies) | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$   |
| $r_{\Delta Ca}$       | annual rate of change for ambient CO <sub>2</sub> concentration                        | $\mu\text{mol mol}^{-1} \text{ yr}^{-1}$     |
| $r_{\Delta Oz}$       | annual rate of change for ground level O <sub>3</sub> concentration                    | $\% \text{ yr}^{-1}$                         |
| $r_{\Delta Pg}$       | annual rate of change for rainfall                                                     | $\text{mm yr}^{-1}$                          |
| $r_{\Delta RH}$       | annual rate of change for relative humidity                                            | $\% \text{ yr}^{-1}$                         |
| $r_{\Delta s}$        | annual rate of change for sunshine hours                                               | $\text{hour yr}^{-1}$                        |
| $r_{\Delta Tmax}$     | annual rate of change for maximum air temperature                                      | $^{\circ}\text{C yr}^{-1}$                   |
| $r_{\Delta Tmin}$     | annual rate of change for minimum air temperature                                      | $^{\circ}\text{C yr}^{-1}$                   |
| $r_{\Delta u}$        | annual rate of change for wind speed                                                   | $\text{m s}^{-1} \text{ yr}^{-1}$            |
| $r_a^a$               | aerodynamic resistance between the mean canopy flow and reference height               | $\text{s m}^{-1}$                            |
| $r_a^c$               | bulk boundary layer resistance                                                         | $\text{s m}^{-1}$                            |
| $r_s^c$               | canopy resistance                                                                      | $\text{s m}^{-1}$                            |
| $r_a^s$               | aerodynamic resistance between the soil surface and mean canopy flow                   | $\text{s m}^{-1}$                            |
| $r_s^s$               | soil surface resistance                                                                | $\text{s m}^{-1}$                            |
| $r_{s,dry}^s$         | soil surface resistance of dry soil                                                    | $\text{s m}^{-1}$                            |
| $P_{g,A}$             | maximum rainfall in the year                                                           | mm                                           |
| $P_{g,d}$             | rainfall (above canopies) for the given day of year                                    | mm                                           |
| $\overline{P_{g,yr}}$ | average rainfall in the year                                                           | mm                                           |

|               |                                                                                     |                          |
|---------------|-------------------------------------------------------------------------------------|--------------------------|
| $P_{net,d}$   | net rainfall (below canopies) for the given day of year                             | mm                       |
| $PAR_{max}$   | maximum PAR                                                                         | $W\ m^{-2}$              |
| $R_{iso}$     | O <sub>3</sub> -forming potential of isoprene (9.1)                                 | $g\ O_3\ g^{-1}\ C_5H_8$ |
| $R_n$         | net solar radiation                                                                 | $W\ m^{-2}$              |
| $R_{nL}$      | net longwave radiation                                                              | $W\ m^{-2}$              |
| $R_{D,e}$     | reduction factor for evaporation                                                    | -                        |
| $R_{D,t}$     | reduction factor for transpiration                                                  | -                        |
| $RH$          | relative humidity for the given hour                                                | %                        |
| $RH_d$        | average relative humidity for the given day of year                                 | %                        |
| $s_d$         | sunshine hours for the given day of year                                            | hour                     |
| $s_i$         | thickness of soil layer $i$                                                         | m                        |
| $S$           | fractional sand content in soil                                                     | -                        |
| $S_i$         | cumulative thickness of soil layer $i$                                              | m                        |
| $SLA$         | specific leaf area                                                                  | $m^2\ g^{-1}$            |
| $t_c$         | the fraction of net radiation as soil heat flux under full canopies (0.05)          | -                        |
| $t_d$         | day of year                                                                         | -                        |
| $t_h$         | local solar time                                                                    | hour                     |
| $t_s$         | the fraction of net radiation as soil heat flux for bare soil (no canopies) (0.315) | -                        |
| $t_{sr}$      | time of sunrise                                                                     | hour                     |
| $t_{ss}$      | time of sunset                                                                      | hour                     |
| $T_a$         | air temperature                                                                     | °C                       |
| $T_a$         | actual plant transpiration                                                          | $m\ day^{-1}$            |
| $T_{avg}$     | average air temperature                                                             | °C                       |
| $T_b$         | base temperature for crop growth                                                    | °C                       |
| $T_f$         | temperature of canopy                                                               | °C                       |
| $T_{a,i}$     | root extraction of water by roots in soil layer $i$                                 | $m\ day^{-1}$            |
| $T_{dew}$     | dew point temperature                                                               | °C                       |
| $T_{dew,cal}$ | calibrated dew point temperature                                                    | °C                       |
| $T_{max}$     | maximum air temperature                                                             | °C                       |

|                                         |                                                                                                                                                                                                                                                                                                                   |                                            |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| $T_{min}$                               | minimum air temperature for the given hour                                                                                                                                                                                                                                                                        | °C                                         |
| $T_{set}$                               | air temperature at sunset                                                                                                                                                                                                                                                                                         | °C                                         |
| $u^*$                                   | friction velocity                                                                                                                                                                                                                                                                                                 | $\text{m s}^{-1}$                          |
| $u$                                     | wind speed at the current hour                                                                                                                                                                                                                                                                                    | $\text{m s}^{-1}$                          |
| $u_A$                                   | highest wind speed in the year                                                                                                                                                                                                                                                                                    | $\text{m s}^{-1}$                          |
| $u_d$                                   | mean wind speed for the given day of year                                                                                                                                                                                                                                                                         | $\text{m s}^{-1}$                          |
| $u_h$                                   | wind speed at canopy height $h$                                                                                                                                                                                                                                                                                   | $\text{m s}^{-1}$                          |
| $u_{max}, u_{min}$                      | maximum and minimum wind speed for the given day, respectively                                                                                                                                                                                                                                                    | $\text{m s}^{-1}$                          |
| $\overline{u_{yr}}$                     | average wind speed in the year                                                                                                                                                                                                                                                                                    | $\text{m s}^{-1}$                          |
| $v_c$                                   | Rubisco-limited rate of CO <sub>2</sub> assimilation                                                                                                                                                                                                                                                              | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ |
| $v_{d,total}, v_{d,canopy}, v_{d,soil}$ | total, canopy, and soil velocity deposition for O <sub>3</sub> , respectively                                                                                                                                                                                                                                     | $\text{m s}^{-1}$                          |
| $v_q$                                   | light-limited rate of CO <sub>2</sub> assimilation                                                                                                                                                                                                                                                                | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ |
| $v_s$                                   | sink-limited rate of CO <sub>2</sub> assimilation                                                                                                                                                                                                                                                                 | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ |
| $V_{c,max}$                             | Rubisco capacity rate (200)                                                                                                                                                                                                                                                                                       | $\mu\text{mol m}^{-2} \text{ leaf s}^{-1}$ |
| $VDM_d$                                 | daily VDM requirement                                                                                                                                                                                                                                                                                             | $\text{kg DM palm}^{-1} \text{ day}^{-1}$  |
| $VDM_{yr}$                              | annual VDM requirement                                                                                                                                                                                                                                                                                            | $\text{kg DM palm}^{-1} \text{ yr}^{-1}$   |
| $VDM_{max,PD}$                          | maximum annual VDM for the given planting density                                                                                                                                                                                                                                                                 | $\text{kg DM palm}^{-1} \text{ yr}^{-1}$   |
| $w$                                     | mean pinnae width                                                                                                                                                                                                                                                                                                 | m                                          |
| $W_x$                                   | dry weight of plant part $x$ , where $x$ is for pinnae, rachis, trunk, roots, male flowers ( <i>maleflo</i> ), immature bunches ( <i>immbunch</i> ), and mature bunches ( <i>matbunch</i> ).<br><br>Note: $W_{trunk}$ is the total of upper ( $W_{top, trunk}$ ) and lower ( $W_{bottom, trunk}$ ) trunk weights. | $\text{kg DM palm}^{-1}$                   |
| $X_c$                                   | correction for mineral content in all plant parts (2.0)                                                                                                                                                                                                                                                           | -                                          |
| $X_x$                                   | fraction by weight of mineral content in plant part $x$ , where $x$ is for pinnae, rachis, trunk, and roots.                                                                                                                                                                                                      | -                                          |

|          |                                                            |   |
|----------|------------------------------------------------------------|---|
| $z_0$    | surface roughness length                                   | m |
| $z_i$    | depth from soil surface to the middle of<br>soil layer $i$ | m |
| $z_r$    | reference height                                           | m |
| $z_{s0}$ | roughness length of soil surface                           | m |

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## APPENDIX B. GROUNDLEVEL OZONE

*Note: This section is still theoretical and lacking in empirical validation.*

Ground level ozone (O<sub>3</sub>) detrimentally affects crop growth and yield. It has been found that increasing levels of O<sub>3</sub> reduces photosynthesis in a linear manner. Interestingly, oil palm is one of the highest crop biogenic emitter of isoprene, which, in turn, reacts with NO<sub>x</sub> (nitrogen oxides) to produce O<sub>3</sub>. In other words, the ambient concentration of ozone above an oil palm land area could potentially be higher than that above a rainforest or a land used for other crops.

### Daily concentration of ozone

The concentration of ozone depends on many factors. Three meteorological properties: wind speed, sunshine hours, and rain are identified as essential to affect the ambient concentration of ground level ozone in Malaysia (Ilyas, 1987). Thus, the annual variation concentration of ozone is determined by

$$O_{z,d} = \overline{O_{z,yr}} + \frac{A \left[ (u_d - \overline{u_{yr}}) / u_A + (s_d - \overline{s_{yr}}) / s_A - (P_{g,d} - \overline{P_{g,yr}}) / P_{g,A} \right]}{3}$$

$$A = \frac{O_{z,yr,max} - O_{z,yr,min}}{O_{z,yr,min}}$$

where  $O_{z,d}$  is the ground level ozone concentration for the day (nmol O<sub>3</sub> mol<sup>-1</sup>);  $\overline{O_{z,yr}}$  is the annual mean ground level ozone concentration (nmol O<sub>3</sub> mol<sup>-1</sup>);  $O_{z,yr,min}$  and  $O_{z,yr,max}$  are the annual minimum and maximum ground level ozone concentration, respectively (nmol O<sub>3</sub> mol<sup>-1</sup>);  $u_d$ ,  $\overline{u_{yr}}$ , and  $u_A$  are the daily wind speed, annual mean wind speed, and annual amplitude of wind speed, respectively (m s<sup>-1</sup>);  $s_d$ ,  $\overline{s_{yr}}$ , and  $s_A$  are the daily sunshine hours, annual mean sunshine hours, and annual amplitude of sunshine hours, respectively (hour); and  $P_{g,d}$ ,  $\overline{P_{g,yr}}$ , and  $P_{g,A}$  are the daily rainfall, annual mean rainfall, and annual amplitude of rainfall, respectively (mm).

### Trend of change in ozone concentration

Ground level ozone is assumed to change in a linear rate per day, using the following equation (model time step equal to one day):

$$\overline{O_{z,yr,(td2)}} = \overline{O_{z,yr,(td1)}} \left( 1 + \frac{r_{\Delta O_z}}{100 \times 365} \right)$$

where  $\overline{O_{z,yr,(td1)}}$  and  $\overline{O_{z,yr,(td2)}}$  are the annual averages of ground level ozone, determined at day of year  $td1$  and  $td2$ , respectively; and  $r_{\Delta O_z}$  is the annual rate of change in ozone



concentration (% yr<sup>-1</sup>). The parameter  $r_{\Delta O_z}$  is taken as a constant, which means that the annual rate of change follows a linear trend.

### Instantaneous concentration of ozone

$$O_z = MAX \left\{ 0, \frac{\pi O_{z,d}}{2} \sin \left[ \frac{\pi}{DL} (t_h - t_{sr}) \right] \right\} \times \frac{N_m}{N_A} \times \frac{1}{1000}$$

where  $O_z$  is the ground level ozone concentration for the hour (converted to  $\mu\text{g O}_3 \text{ m}^{-3}$ );  $O_{z,d}$  is the ground level ozone concentration for the day ( $\text{nmol O}_3 \text{ mol}^{-1}$ );  $t_h$  and  $t_{sr}$  are the current hour and hour of sunrise, respectively (hour);  $DL$  is the day length (hour); and  $N_m$  and  $N_A$  are the number of molecules in per cubic meter of air and in per mole of air, respectively.

*Note:*  $N_m$  is  $2.7 \times 10^{25}$  molecule  $\text{m}^{-3}$  air, and  $N_A$  (Avogadro number) is  $6.022 \times 10^{23}$  molecule  $\text{mol}^{-1}$  air.

### Isoprene emission

Isoprene ( $\text{C}_5\text{H}_8$ ) emitted by oil palm leaves is a function of both solar irradiance and canopy temperature. The bet fit function was fitted to the measured isoprene emission by oil palm leaves (Wilkinson et al, 2006). For greater accuracy, isoprene emission is calculated separately for sunlit and shaded leaves as

$$E_{iso,sl/sh} = \exp \left( a + \frac{b}{I_{sl/sh}^2} + \frac{c}{T_f^2} \right) \times f_{iso} \times \frac{SLA}{1000} \times \frac{68}{12} \times 12 \times 10^{-9}$$

$$a = 4.4613$$

$$b = -23781.5312$$

$$c = -1597.2599$$

where  $E_{iso,sl/sh}$  is the emitted isoprene concentration (converted to unit  $\text{g C}_5\text{H}_8 \text{ g}^{-1} \text{ leaf s}^{-1}$ ) by sunlit (subscript *sl*) or shaded (subscript *sh*) leaves;  $I_{sl/sh}$  is the incident solar irradiance ( $\mu\text{mol photons m}^{-2} \text{ ground s}^{-1}$ ) on sunlit (*sl*) or shaded (*sh*) leaves;  $T_f$  is the canopy temperature ( $^{\circ}\text{C}$ );  $SLA$  is the specific leaf area ( $\text{m}^2 \text{ leaf kg}^{-1} \text{ leaf}$ ); and  $f_{iso}$  is the  $\text{CO}_2$ -inhibited isoprene factor (unitless).  $f_{iso}$  is calculated by fitting the best function to the collected measured data set from Young *et al.* (2009):

$$f_{iso} = p + \frac{q \ln C_a}{C_a^2}$$

$$p = 0.489881$$

$$q = 11231.78$$

where  $C_a$  is the ambient  $\text{CO}_2$  concentration ( $\mu\text{mol mol}^{-1}$ ). Values of  $f_{iso}$  less than 1 mean that isoprene emission is inhibited by high levels of  $\text{CO}_2$ . In contrast, values of  $f_{iso}$  greater than 1 mean that isoprene emission is encouraged by low levels of  $\text{CO}_2$ .

### Ozone created from isoprene emission

$$O_{z,sl} = E_{iso,sl} \times R_{iso} \times \frac{L_{sl}}{SLA} \times \frac{10000}{PD} \times 1000$$

$$O_{z,sh} = E_{iso,sh} \times R_{iso} \times \frac{L_{sh}}{SLA} \times \frac{10000}{PD} \times 1000$$

where  $O_{z,sl}$  and  $O_{z,sh}$  are the ozone created from isoprene emission from sunlit and shaded leaves, respectively (converted to  $\text{g O}_3 \text{ palm}^{-1} \text{ s}^{-1}$ );  $E_{iso,sl}$  and  $E_{iso,sh}$  are the emitted isoprene concentration ( $\text{g C}_5\text{H}_8 \text{ g}^{-1} \text{ leaf s}^{-1}$ ) by sunlit and shaded leaves, respectively;  $SLA$  is the specific leaf area ( $\text{m}^2 \text{ leaf kg}^{-1} \text{ leaf}$ );  $PD$  is the planting density ( $\text{palms ha}^{-1}$ );  $L_{sl}$  and  $L_{sh}$  are sunlit and shaded leaf area index, respectively ( $\text{m}^2 \text{ leaf m}^{-2} \text{ ground}$ ); and  $R_{iso}$  is the ozone-forming potential ( $\text{g O}_3 \text{ g}^{-1} \text{ C}_5\text{H}_8$ ), determined to be equal to  $9.1 \text{ g O}_3 \text{ g}^{-1} \text{ C}_5\text{H}_8$  (Benjamin and Winer, 1998).

Consequently, total ozone production is

$$O_{z,prod} = (O_{z,sl} + O_{z,sh}) \times \frac{PD}{10000} \times 10^6$$

where  $O_{z,prod}$  is the total ozone concentration (background  $\text{O}_3$  and  $\text{O}_3$  produced from isoprene emission by oil palm leaves) (converted to  $\mu\text{g O}_3 \text{ m}^{-2} \text{ ground s}^{-1}$ );  $O_{z,sl}$  and  $O_{z,sh}$  are the ozone created from isoprene emission from sunlit and shaded leaves, respectively ( $\text{g O}_3 \text{ palm}^{-1} \text{ s}^{-1}$ ); and  $PD$  is the planting density ( $\text{palms ha}^{-1}$ ).

### Dry deposition velocity

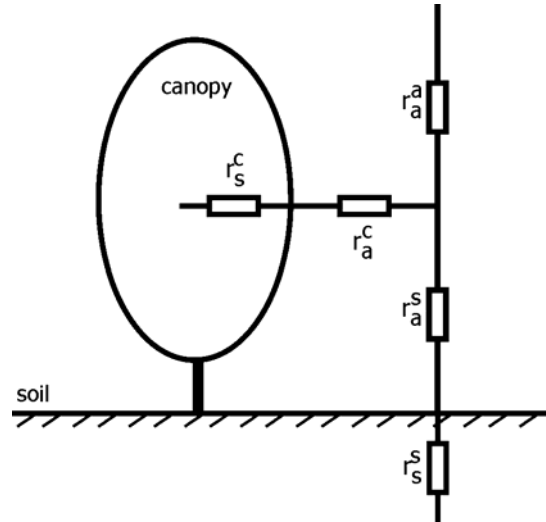


Fig. B.1. Schematic network of resistances to determine the ozone deposition velocities of canopy and soil. Key:  $r_a^a$  is the aerodynamic resistance between the canopy and reference height;  $r_a^s$  is the aerodynamic resistance between the soil and canopy;  $r_a^c$  is the bulk boundary layer resistance;  $r_s^c$  and  $r_s^s$  are the canopy and soil surface resistance, respectively.

Ozone dry deposition occurs on both canopy and soil. Their deposition velocities are described by

$$v_{d,total} = v_{d,canopy} + v_{d,soil}$$

$$v_{d,canopy} = [R_T (1 + R_c / R_s)]^{-1}$$

$$v_{d,soil} = [R_T (1 + R_s / R_c)]^{-1}$$

$$R_T = r_a^a + \left( \frac{1}{R_c} + \frac{1}{R_a} \right)^{-1}$$

$$R_c = r_a^c + 1.67 r_s^c$$

$$R_s = r_a^s + 1.67 r_s^s$$

where  $v_{d,canopy}$  and  $v_{d,soil}$  are the canopy and soil deposition velocities, respectively ( $\text{m s}^{-1}$ );  $v_{d,total}$  is the total deposition velocity ( $\text{m s}^{-1}$ );  $r_a^a$  is the aerodynamic resistance between the canopy and reference height ( $\text{s m}^{-1}$ );  $r_a^s$  is the aerodynamic resistance between the soil and canopy ( $\text{s m}^{-1}$ );  $r_a^c$  is the bulk boundary layer resistance ( $\text{s m}^{-1}$ );  $r_s^c$  and  $r_s^s$  are the canopy and soil surface resistance, respectively ( $\text{s m}^{-1}$ ).

### Net ozone concentration

The ambient concentration of ozone is calculated based on the Eulerian box model for atmospheric transport (Seinfeld and Pandis, 2006):

$$O_{z,net} = \frac{1}{48} \times \frac{A}{B} [1 - \exp(-B \times t_h \times 3600)] \times 0.5$$

where  $O_{z,net}$  is the net concentration of ozone (converted to  $\mu\text{mol O}_3 \text{ m}^{-3}$ );  $t_h$  is the local solar hour (hour); and coefficients  $A$  and  $B$  are determined by

$$A = \frac{O_{z,prod}}{l} + \frac{O_z}{\tau}$$

$$B = O_{z,loss} + \frac{v_{d,total}}{l} + \frac{1}{\tau}$$

$$\tau = \frac{l}{u}$$

$$O_{z,loss} = \frac{1}{12 \times 60 \times 60}$$

where  $O_{z,prod}$  is the total ozone concentration (background  $O_3$  and  $O_3$  produced from isoprene emission by oil palm leaves) ( $\mu\text{g } O_3 \text{ m}^{-2} \text{ ground s}^{-1}$ );  $O_z$  is the ground level ozone concentration for the hour ( $\mu\text{g } O_3 \text{ m}^{-3}$ );  $v_{d,total}$  is the total deposition velocity ( $\text{m s}^{-1}$ );  $l$  is the mixing height (m), taken as 1000 m;  $u$  is the wind speed for the hour ( $\text{m s}^{-1}$ );  $\tau$  is the ozone residence time (s); and  $O_{z,loss}$  is the rate of  $O_3$  chemical reaction in ambient air ( $\text{s}^{-1}$ ). It is assumed that within 12 hours (or  $12 \times 60 \times 60$  s), all ground level  $O_3$  would have reacted completely.

*Note:* Multiplication by 0.5 to obtain net ozone concentration at the ground level is done assuming ozone concentration near ground is less by half of that in the upper atmosphere.

### Reduction of photosynthesis

It has been measured that ozone reduces photosynthesis in a linear manner (Reich, 1987). To determine ozone damage, it is necessary to first determine the ozone flux into the leaves; that is, to determine the ozone deposition on leaves, determined by

$$F_{stom*} = O_{z,net} \times v_{d,canopy}$$

where  $F_{stom*}$  is the crop's instantaneous ozone dose ( $\mu\text{mol } O_3 \text{ m}^{-2} \text{ ground s}^{-1}$ );  $O_{z,net}$  is the net concentration of ozone ( $\mu\text{mol } O_3 \text{ m}^{-3}$ ); and  $v_{d,canopy}$  is the ozone deposition velocity to the canopy ( $\text{m s}^{-1}$ ).

Integrating the instantaneous ozone dose over a given period gives the ozone effective dose, or

$$F_{O3eff} = \frac{3600}{1000} \int_0^t F_{stom*} dt'$$

where  $F_{O3eff}$  is the effective ozone dose (converted to  $\text{mmol } O_3 \text{ m}^{-2} \text{ ground}$ ). The amount of assimilates reduced due to ozone is then determined by

$$f_{O3} = 1.946 \times 10^{-4} \times F_{O3eff}$$

where  $f_{O3}$  is the reduction in assimilates (fraction); and  $1.946 \times 10^{-4}$  is the measured slope between the fraction of photosynthesis reduction and the effective ozone dose for hardwoods (*i.e.*, angiosperm trees like oil palm) (Reich, 1987).